Compute Cell Scheduling Records

This document records how tasks were scheduled on small compute clusters. The workload and the machine failures are real records from a production compute cell of about 12,500 machines (the Google cluster trace of May 2011, clusterdata-2011-2, CC-BY 4.0): every task's submission time, job, priority, CPU and memory request and run time, and every machine's capacity, failure time and return time, are taken from that trace (ids are renamed). Where and when each task ran in the small cluster is not recorded in the trace: it is computed from those records by the scheduling policy stated below, applied exactly.

The document is divided into chapters titled 'Cell window <NN>'. Each chapter is an independent run of one small cluster over a window of 1200 seconds. Questions refer to a chapter by that title. Times are whole seconds from the start of the window (time 0); the window covers times 0 to 1200.

Each chapter has four sections. 'Machines' lists the cluster's machines with their CPU and memory capacity, the time a machine failed (if it did) and the time it returned (if it did). 'Task log' lists, in time order, every event of every task from time 0 to the window end: submit, start, finish, evict and fail, with the task's job, priority, CPU and memory request, run time, and the machine involved. 'Events' lists machine failures and returns, evictions and restarts, and the longest waits. 'Queue at window end' gives the state after the events at time 1200: the tasks still running (with their remaining run time) and the tasks still waiting, in queue order.

Tasks. A task belongs to a job and has a priority from 0 to 11 (higher is more important). Priorities 9 to 11 are the *production* tier and priorities 0 and 1 the *free* tier. Each task requests an amount of CPU and of memory, which it holds on one machine while it runs, and has a run time: once started it runs for exactly that many seconds and then finishes, unless it is stopped first. The run time is given on every row of the task log, starting with the task's submit row.

Scheduling policy. Every window starts with all machines empty and no task waiting. At every time at which something happens, in this order:

1. **Finishes.** Tasks whose run time is over finish and free their CPU and memory.

2. **Failures.** Machines that fail at this time go down. Each task running on such a machine stops (a *fail* event) and goes back to the queue.

3. **Returns.** Machines that return at this time come back, empty, with their capacity.

4. **Submissions.** Tasks submitted at this time join the queue.

5. **Scheduling round.** The waiting tasks are taken one at a time in *queue order*: higher priority first, then earlier submission time, then lower task id. Each task, in turn:

a. **First fit.** The task starts on the first machine, in machine id order, that is up and whose free CPU and free memory are both at least the task's requests (subject to the reservation rule in c).

b. **Preemption.** If it fits nowhere, no task holds a reservation yet in this round, and it is a production task, it may evict free-tier tasks from one machine. It looks at the up machines in id order. On each it considers the free-tier tasks running there, lowest priority first, then the most recently started first (then the higher task id first), and evicts them one by one in that order until it fits; it takes the first machine where this works, evicting only as many as needed there, and starts on it. If no machine works, nothing is evicted. An evicted task goes back to the queue at once (it may start again later in the same round); it loses its progress and will run its full run time again.

c. **Reservation and backfilling.** The first task in the round that can start neither by a. nor by b. gets a *reservation*: the earliest time at which it could start, computed from the run times of the tasks now running (assuming no new submissions, failures or returns), on the first machine, in id order, where that earliest time is reached. Every later task in the round may still start by first fit, but on the reserved machine only if it will finish no later than the reserved time, or if its requests fit in what the reserved machine will still have free at the reserved time after the reserved task's requests (and after other such starts on it in the same round). On the other machines it starts as usual. Later tasks do not preempt. A task that would not fit on any up machine even if that machine were empty gets no reservation and simply keeps waiting; the next task that cannot start may then get the reservation.

Tasks stopped by an eviction or a failure keep their original submission time (and so their place in queue order) and start again from the beginning. A task never moves while it runs.

Order of rows. Task log rows with the same time follow the steps above: finishes (by task id), failures (by machine, then task id), submissions (by task id), then the scheduling round's starts and evictions in the order the round makes them (each eviction just before the start that caused it).

Terms used in the Events section and in questions. A task's *wait* is the time from its submission to its first start. A task is *restarted* when it starts again after an eviction or a failure. The number of *evictions* counts evict events (one per evicted task). 'Running on a machine at time t' means running after all the events at time t.

The cluster and the workload. Each window has 8 machines, ids M01 to M08. Each is modelled on a real machine of the cell (one of them is a machine that failed during the window in the trace) and has 1/8 of that machine's recorded CPU and memory capacity. A failed machine returns at the time the trace records it coming back, if that is within 6 hours; otherwise it does not return. The tasks are a sample of the real tasks submitted in the cell during the window, chosen by this fixed rule: jobs submitted in the window are taken in a random order, first up to 6 production jobs, then free-tier jobs until the offered load reaches 54%, then other jobs until it reaches 90% (a job that would push it above 95% is skipped). A job contributes its first 4 tasks (16 for a free-tier job) that finished normally in the trace and fit on at least one machine of the window. Offered load = the larger, over CPU and memory, of the sum of request x run time inside the window over all tasks, divided by the cluster's total capacity x the window length.

After the window. For questions about what happens after time 1200: no new task is submitted and no machine fails after the window; a machine that is down returns at its listed return time (if any); the policy continues unchanged from the state given in 'Queue at window end'.

Units. Times and run times in seconds. CPU and memory in units of 1/100,000 of the largest machine in the original cell (the trace's normalised values, rounded to whole units).

Chapters in this document (itops_d04): Cell window 02, Cell window 03, Cell window 20, Cell window 23, Cell window 26.

Cell window 02

Times 0 to 1200. The cluster has 8 machines. 103 tasks of 55 jobs are submitted during the window.

Cell window 02: Machines

machine	CPU	memory	fails at	returns at
M01	6250	6244	389	992
M02	12500	12500	-	-
M03	12500	12500	-	-
M04	3125	3123	-	-
M05	6250	6244	-	-
M06	6250	6244	-	-
M07	6250	3116	-	-
M08	6250	6244	-	-

Cell window 02: Task log

time	task	job	priority	CPU	memory	run time	event	machine
0	T001	J01	4	2499	3339	389	submit	-
0	T002	J01	4	2499	3339	346	submit	-
0	T003	J01	4	2499	3339	155	submit	-
0	T004	J01	4	2499	3339	233	submit	-
0	T001	J01	4	2499	3339	389	start	M01
0	T002	J01	4	2499	3339	346	start	M02
0	T003	J01	4	2499	3339	155	start	M02
0	T004	J01	4	2499	3339	233	start	M02
15	T005	J02	4	6250	1553	64	submit	-
15	T006	J02	4	6250	1553	52	submit	-
15	T007	J02	4	6250	1553	118	submit	-
15	T008	J02	4	6250	1553	80	submit	-
15	T009	J03	8	6250	1553	323	submit	-
15	T009	J03	8	6250	1553	323	start	M03
15	T005	J02	4	6250	1553	64	start	M03
15	T006	J02	4	6250	1553	52	start	M05
15	T007	J02	4	6250	1553	118	start	M06
15	T008	J02	4	6250	1553	80	start	M07
17	T010	J04	4	6250	1553	75	submit	-
17	T011	J04	4	6250	1553	74	submit	-
17	T012	J04	4	6250	1553	64	submit	-
17	T013	J04	4	6250	1553	65	submit	-
17	T010	J04	4	6250	1553	75	start	M08
33	T014	J05	8	6250	932	1099	submit	-
36	T015	J06	0	1250	16	24	submit	-
36	T015	J06	0	1250	16	24	start	M01
52	T016	J07	4	1250	398	44	submit	-
52	T016	J07	4	1250	398	44	start	M01
60	T015	J06	0	1250	16	24	finish	M01
67	T006	J02	4	6250	1553	52	finish	M05
67	T014	J05	8	6250	932	1099	start	M05
75	T017	J08	8	2499	3339	897	submit	-
79	T005	J02	4	6250	1553	64	finish	M03
79	T017	J08	8	2499	3339	897	start	M03
92	T010	J04	4	6250	1553	75	finish	M08
92	T011	J04	4	6250	1553	74	start	M08
93	T018	J09	4	1250	466	216	submit	-
93	T018	J09	4	1250	466	216	start	M01

time	task	job	priority	CPU	memory	run time	event	machine
95	T008	J02	4	6250	1553	80	finish	M07
95	T012	J04	4	6250	1553	64	start	M07
96	T016	J07	4	1250	398	44	finish	M01
110	T019	J10	4	1250	466	182	submit	-
110	T020	J10	4	1250	466	182	submit	-
110	T021	J10	4	1250	466	193	submit	-
110	T022	J10	4	1250	466	178	submit	-
110	T023	J11	8	6250	4773	843	submit	-
110	T019	J10	4	1250	466	182	start	M01
110	T020	J10	4	1250	466	182	start	M01
110	T021	J10	4	1250	466	193	start	M02
110	T022	J10	4	1250	466	178	start	M02
133	T007	J02	4	6250	1553	118	finish	M06
133	T023	J11	8	6250	4773	843	start	M06
152	T024	J12	4	1250	3107	1540	submit	-
152	T025	J12	4	1250	3107	1539	submit	-
152	T026	J12	4	1250	3107	1540	submit	-
152	T027	J12	4	1250	3107	99	submit	-
152	T024	J12	4	1250	3107	1540	start	M03
152	T025	J12	4	1250	3107	1539	start	M03
152	T026	J12	4	1250	3107	1540	start	M04
155	T003	J01	4	2499	3339	155	finish	M02
155	T027	J12	4	1250	3107	99	start	M02
159	T012	J04	4	6250	1553	64	finish	M07
159	T013	J04	4	6250	1553	65	start	M07
166	T011	J04	4	6250	1553	74	finish	M08
169	T028	J13	4	3125	1125	46	submit	-
169	T029	J14	4	1250	466	209	submit	-
169	T028	J13	4	3125	1125	46	start	M02
169	T029	J14	4	1250	466	209	start	M03
185	T030	J15	4	1250	466	189	submit	-
185	T031	J15	4	1250	466	186	submit	-
185	T032	J15	4	1250	466	184	submit	-
185	T033	J15	4	1250	466	188	submit	-
185	T030	J15	4	1250	466	189	start	M08
185	T031	J15	4	1250	466	186	start	M08
185	T032	J15	4	1250	466	184	start	M08
185	T033	J15	4	1250	466	188	start	M08
215	T028	J13	4	3125	1125	46	finish	M02
224	T013	J04	4	6250	1553	65	finish	M07
233	T004	J01	4	2499	3339	233	finish	M02
254	T027	J12	4	1250	3107	99	finish	M02
288	T022	J10	4	1250	466	178	finish	M02
292	T019	J10	4	1250	466	182	finish	M01
292	T020	J10	4	1250	466	182	finish	M01
303	T021	J10	4	1250	466	193	finish	M02
309	T018	J09	4	1250	466	216	finish	M01
336	T034	J16	1	3125	795	6831	submit	-
336	T035	J17	0	6250	466	45	submit	-
336	T034	J16	1	3125	795	6831	start	M01
336	T035	J17	0	6250	466	45	start	M02
338	T009	J03	8	6250	1553	323	finish	M03
339	T036	J18	0	3125	795	6827	submit	-
339	T036	J18	0	3125	795	6827	start	M02
342	T037	J19	0	625	387	91	submit	-
342	T038	J19	0	625	387	92	submit	-
342	T039	J19	0	625	387	93	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
342	T040	J19	0	625	387	102	submit	-
342	T041	J19	0	625	387	90	submit	-
342	T042	J19	0	625	387	92	submit	-
342	T043	J19	0	625	387	98	submit	-
342	T044	J19	0	625	387	99	submit	-
342	T045	J19	0	625	387	89	submit	-
342	T046	J19	0	625	387	188	submit	-
342	T047	J19	0	625	387	118	submit	-
342	T048	J19	0	625	387	112	submit	-
342	T049	J19	0	625	387	112	submit	-
342	T050	J19	0	625	387	93	submit	-
342	T051	J19	0	625	387	90	submit	-
342	T052	J19	0	625	387	108	submit	-
342	T037	J19	0	625	387	91	start	M01
342	T038	J19	0	625	387	92	start	M02
342	T039	J19	0	625	387	93	start	M03
342	T040	J19	0	625	387	102	start	M03
342	T041	J19	0	625	387	90	start	M03
342	T042	J19	0	625	387	92	start	M03
342	T043	J19	0	625	387	98	start	M03
342	T044	J19	0	625	387	99	start	M03
342	T045	J19	0	625	387	89	start	M07
342	T046	J19	0	625	387	188	start	M07
342	T047	J19	0	625	387	118	start	M07
342	T048	J19	0	625	387	112	start	M07
342	T049	J19	0	625	387	112	start	M07
342	T050	J19	0	625	387	93	start	M07
342	T051	J19	0	625	387	90	start	M07
342	T052	J19	0	625	387	108	start	M07
344	T053	J20	4	625	155	40	submit	-
344	T053	J20	4	625	155	40	start	M03
346	T002	J01	4	2499	3339	346	finish	M02
364	T054	J21	4	625	155	32	submit	-
364	T054	J21	4	625	155	32	start	M02
365	T055	J22	4	625	311	28	submit	-
365	T055	J22	4	625	311	28	start	M02
369	T032	J15	4	1250	466	184	finish	M08
371	T031	J15	4	1250	466	186	finish	M08
373	T033	J15	4	1250	466	188	finish	M08
374	T030	J15	4	1250	466	189	finish	M08
378	T029	J14	4	1250	466	209	finish	M03
381	T035	J17	0	6250	466	45	finish	M02
383	T056	J23	4	6250	1553	291	submit	-
383	T057	J24	4	6250	1553	96	submit	-
383	T058	J24	4	6250	1553	72	submit	-
383	T059	J24	4	6250	1553	59	submit	-
383	T060	J24	4	6250	1553	67	submit	-
383	T056	J23	4	6250	1553	291	start	M02
383	T057	J24	4	6250	1553	96	start	M08
384	T053	J20	4	625	155	40	finish	M03
389	T001	J01	4	2499	3339	389	finish	M01
389	T034	J16	1	3125	795	6831	fail	M01
389	T037	J19	0	625	387	91	fail	M01
389	T037	J19	0	625	387	91	start	M02
393	T055	J22	4	625	311	28	finish	M02
396	T054	J21	4	625	155	32	finish	M02
431	T045	J19	0	625	387	89	finish	M07

time	task	job	priority	CPU	memory	run time	event	machine
432	T041	J19	0	625	387	90	finish	M03
432	T051	J19	0	625	387	90	finish	M07
434	T038	J19	0	625	387	92	finish	M02
434	T042	J19	0	625	387	92	finish	M03
435	T039	J19	0	625	387	93	finish	M03
435	T050	J19	0	625	387	93	finish	M07
435	T034	J16	1	3125	795	6831	start	M07
440	T043	J19	0	625	387	98	finish	M03
440	T058	J24	4	6250	1553	72	start	M03
441	T044	J19	0	625	387	99	finish	M03
444	T040	J19	0	625	387	102	finish	M03
450	T052	J19	0	625	387	108	finish	M07
454	T048	J19	0	625	387	112	finish	M07
454	T049	J19	0	625	387	112	finish	M07
456	T061	J25	0	6250	1244	149	submit	-
457	T062	J26	0	6250	1244	165	submit	-
460	T047	J19	0	625	387	118	finish	M07
463	T063	J27	4	62	398	16	submit	-
463	T063	J27	4	62	398	16	start	M02
479	T057	J24	4	6250	1553	96	finish	M08
479	T063	J27	4	62	398	16	finish	M02
479	T059	J24	4	6250	1553	59	start	M08
480	T037	J19	0	625	387	91	finish	M02
512	T058	J24	4	6250	1553	72	finish	M03
512	T060	J24	4	6250	1553	67	start	M03
516	T064	J28	1	62	398	24	submit	-
516	T064	J28	1	62	398	24	start	M02
517	T065	J29	9	6250	1590	74	submit	-
517	T036	J18	0	3125	795	6827	evict	M02
517	T064	J28	1	62	398	24	evict	M02
517	T065	J29	9	6250	1590	74	start	M02
517	T064	J28	1	62	398	24	start	M03
530	T046	J19	0	625	387	188	finish	M07
530	T036	J18	0	3125	795	6827	start	M07
538	T059	J24	4	6250	1553	59	finish	M08
538	T061	J25	0	6250	1244	149	start	M08
541	T064	J28	1	62	398	24	finish	M03
576	T066	J30	0	3125	311	330	submit	-
579	T060	J24	4	6250	1553	67	finish	M03
579	T062	J26	0	6250	1244	165	start	M03
591	T065	J29	9	6250	1590	74	finish	M02
591	T066	J30	0	3125	311	330	start	M02
636	T067	J31	9	6250	3180	214	submit	-
636	T066	J30	0	3125	311	330	evict	M02
636	T067	J31	9	6250	3180	214	start	M02
674	T056	J23	4	6250	1553	291	finish	M02
674	T066	J30	0	3125	311	330	start	M02
687	T061	J25	0	6250	1244	149	finish	M08
744	T062	J26	0	6250	1244	165	finish	M03
746	T068	J32	4	1250	466	180	submit	-
746	T068	J32	4	1250	466	180	start	M02
747	T069	J33	4	1250	466	178	submit	-
747	T070	J33	4	1250	466	177	submit	-
747	T071	J33	4	1250	466	179	submit	-
747	T072	J33	4	1250	466	177	submit	-
747	T069	J33	4	1250	466	178	start	M02
747	T070	J33	4	1250	466	177	start	M03

time	task	job	priority	CPU	memory	run time	event	machine
747	T071	J33	4	1250	466	179	start	M03
747	T072	J33	4	1250	466	177	start	M03
850	T067	J31	9	6250	3180	214	finish	M02
924	T070	J33	4	1250	466	177	finish	M03
924	T072	J33	4	1250	466	177	finish	M03
925	T069	J33	4	1250	466	178	finish	M02
926	T068	J32	4	1250	466	180	finish	M02
926	T071	J33	4	1250	466	179	finish	M03
926	T073	J34	8	2499	3339	1159	submit	-
926	T073	J34	8	2499	3339	1159	start	M02
936	T074	J35	0	6250	466	32	submit	-
936	T074	J35	0	6250	466	32	start	M02
939	T075	J36	9	6250	3180	214	submit	-
939	T076	J37	9	4688	6360	521	submit	-
939	T077	J38	1	1250	1590	309	submit	-
939	T075	J36	9	6250	3180	214	start	M08
939	T074	J35	0	6250	466	32	evict	M02
939	T076	J37	9	4688	6360	521	start	M02
939	T077	J38	1	1250	1590	309	start	M02
939	T074	J35	0	6250	466	32	start	M03
940	T078	J39	9	6250	10340	587	submit	-
943	T079	J40	8	6250	4773	569	submit	-
962	T080	J41	4	2499	3339	177	submit	-
962	T081	J41	4	2499	3339	227	submit	-
962	T082	J41	4	2499	3339	186	submit	-
962	T083	J41	4	2499	3339	249	submit	-
962	T084	J42	4	1250	466	206	submit	-
962	T085	J43	4	3125	1125	30	submit	-
962	T086	J44	8	3125	1840	353	submit	-
962	T084	J42	4	1250	466	206	start	M03
971	T074	J35	0	6250	466	32	finish	M03
971	T086	J44	8	3125	1840	353	start	M03
976	T017	J08	8	2499	3339	897	finish	M03
976	T023	J11	8	6250	4773	843	finish	M06
976	T079	J40	8	6250	4773	569	start	M06
976	T080	J41	4	2499	3339	177	start	M03
992	T081	J41	4	2499	3339	227	start	M01
992	T085	J43	4	3125	1125	30	start	M01
999	T087	J45	0	6250	1244	468	submit	-
1000	T088	J46	0	6250	1244	120	submit	-
1001	T089	J47	0	6250	1244	144	submit	-
1004	T066	J30	0	3125	311	330	finish	M02
1022	T085	J43	4	3125	1125	30	finish	M01
1043	T090	J48	8	6250	1553	546	submit	-
1059	T091	J49	8	6250	4773	1001	submit	-
1061	T092	J50	4	2499	3339	308	submit	-
1061	T093	J50	4	2499	3339	308	submit	-
1061	T094	J50	4	2499	3339	335	submit	-
1061	T095	J50	4	2499	3339	290	submit	-
1101	T096	J51	4	6250	1553	59	submit	-
1101	T097	J51	4	6250	1553	73	submit	-
1101	T098	J51	4	6250	1553	70	submit	-
1101	T099	J51	4	6250	1553	62	submit	-
1136	T100	J52	8	1250	2863	1411	submit	-
1136	T100	J52	8	1250	2863	1411	start	M01
1153	T075	J36	9	6250	3180	214	finish	M08
1153	T080	J41	4	2499	3339	177	finish	M03

time	task	job	priority	CPU	memory	run time	event	machine
1153	T101	J53	4	1250	466	232	submit	-
1153	T090	J48	8	6250	1553	546	start	M08
1153	T082	J41	4	2499	3339	186	start	M03
1153	T101	J53	4	1250	466	232	start	M02
1166	T014	J05	8	6250	932	1099	finish	M05
1166	T091	J49	8	6250	4773	1001	start	M05
1168	T084	J42	4	1250	466	206	finish	M03
1176	T102	J54	0	6250	1244	223	submit	-
1177	T103	J55	0	6250	1244	228	submit	-

Cell window 02: Events

Time 389: machine M01 failed; tasks T034, T037 stopped and went back to the queue.

Time 389: task T037 restarted on machine M02.

Time 435: task T034 restarted on machine M07.

Time 517: task T065 (priority 9) evicted task T036 (priority 0) from machine M02.

Time 517: task T065 (priority 9) evicted task T064 (priority 1) from machine M02.

Time 517: task T064 restarted on machine M03.

Time 530: task T036 restarted on machine M07.

Time 636: task T067 (priority 9) evicted task T066 (priority 0) from machine M02.

Time 674: task T066 restarted on machine M02.

Time 939: task T076 (priority 9) evicted task T074 (priority 0) from machine M02.

Time 939: task T074 restarted on machine M03.

Time 992: machine M01 returned.

Longest waits (submission to first start, tasks started by the window end): task T082 waited 191 s (submitted at 962, first started at 1153); task T013 waited 142 s (submitted at 17, first started at 159); task T060 waited 129 s (submitted at 383, first started at 512); task T062 waited 122 s (submitted at 457, first started at 579); task T090 waited 110 s (submitted at 1043, first started at 1153).

Cell window 02: Queue at window end

State after the events at time 1200.

Tasks running at time 1200 (remaining run time = seconds still to run after time 1200):

task	job	priority	CPU	memory	machine	started	remaining
T081	J41	4	2499	3339	M01	992	19
T100	J52	8	1250	2863	M01	1136	1347
T073	J34	8	2499	3339	M02	926	885
T076	J37	9	4688	6360	M02	939	260
T077	J38	1	1250	1590	M02	939	48
T101	J53	4	1250	466	M02	1153	185
T024	J12	4	1250	3107	M03	152	492
T025	J12	4	1250	3107	M03	152	491
T082	J41	4	2499	3339	M03	1153	139
T086	J44	8	3125	1840	M03	971	124
T026	J12	4	1250	3107	M04	152	492
T091	J49	8	6250	4773	M05	1166	967
T079	J40	8	6250	4773	M06	976	345
T034	J16	1	3125	795	M07	435	6066
T036	J18	0	3125	795	M07	530	6157

task	job	priority	CPU	memory	machine	started	remaining
T090	J48	8	6250	1553	M08	1153	499

Tasks waiting at time 1200, in queue order (each will run its full run time once started):

task	job	priority	CPU	memory	submitted	run time
T078	J39	9	6250	10340	940	587
T083	J41	4	2499	3339	962	249
T092	J50	4	2499	3339	1061	308
T093	J50	4	2499	3339	1061	308
T094	J50	4	2499	3339	1061	335
T095	J50	4	2499	3339	1061	290
T096	J51	4	6250	1553	1101	59
T097	J51	4	6250	1553	1101	73
T098	J51	4	6250	1553	1101	70
T099	J51	4	6250	1553	1101	62
T087	J45	0	6250	1244	999	468
T088	J46	0	6250	1244	1000	120
T089	J47	0	6250	1244	1001	144
T102	J54	0	6250	1244	1176	223
T103	J55	0	6250	1244	1177	228

All machines are up at time 1200.

Cell window 03

Times 0 to 1200. The cluster has 8 machines. 123 tasks of 52 jobs are submitted during the window.

Cell window 03: Machines

machine	CPU	memory	fails at	returns at
M01	6250	1551	-	-
M02	6250	9363	-	-
M03	6250	6244	-	-
M04	6250	6244	-	-
M05	6250	3116	-	-
M06	6250	3116	-	-
M07	6250	6244	541	1505
M08	6250	6244	-	-

Cell window 03: Task log

time	task	job	priority	CPU	memory	run time	event	machine
48	T001	J01	4	62	398	447	submit	-
48	T001	J01	4	62	398	447	start	M01
94	T002	J02	1	3125	795	6831	submit	-
94	T003	J03	0	6250	466	45	submit	-
94	T002	J02	1	3125	795	6831	start	M01
94	T003	J03	0	6250	466	45	start	M02
97	T004	J04	0	3125	795	6827	submit	-
97	T004	J04	0	3125	795	6827	start	M03
100	T005	J05	0	625	387	91	submit	-
100	T006	J05	0	625	387	92	submit	-
100	T007	J05	0	625	387	93	submit	-
100	T008	J05	0	625	387	102	submit	-
100	T009	J05	0	625	387	90	submit	-
100	T010	J05	0	625	387	92	submit	-
100	T011	J05	0	625	387	98	submit	-
100	T012	J05	0	625	387	99	submit	-
100	T013	J05	0	625	387	89	submit	-
100	T014	J05	0	625	387	188	submit	-
100	T015	J05	0	625	387	118	submit	-
100	T016	J05	0	625	387	112	submit	-
100	T017	J05	0	625	387	112	submit	-
100	T018	J05	0	625	387	93	submit	-
100	T019	J05	0	625	387	90	submit	-
100	T020	J05	0	625	387	108	submit	-
100	T005	J05	0	625	387	91	start	M03
100	T006	J05	0	625	387	92	start	M03
100	T007	J05	0	625	387	93	start	M03
100	T008	J05	0	625	387	102	start	M03
100	T009	J05	0	625	387	90	start	M03
100	T010	J05	0	625	387	92	start	M04
100	T011	J05	0	625	387	98	start	M04
100	T012	J05	0	625	387	99	start	M04
100	T013	J05	0	625	387	89	start	M04
100	T014	J05	0	625	387	188	start	M04
100	T015	J05	0	625	387	118	start	M04
100	T016	J05	0	625	387	112	start	M04
100	T017	J05	0	625	387	112	start	M04
100	T018	J05	0	625	387	93	start	M04

time	task	job	priority	CPU	memory	run time	event	machine
100	T019	J05	0	625	387	90	start	M04
100	T020	J05	0	625	387	108	start	M05
102	T021	J06	8	6250	1553	347	submit	-
102	T021	J06	8	6250	1553	347	start	M06
123	T022	J07	4	6250	1553	327	submit	-
123	T023	J08	4	6250	1553	62	submit	-
123	T024	J08	4	6250	1553	91	submit	-
123	T025	J08	4	6250	1553	62	submit	-
123	T026	J08	4	6250	1553	132	submit	-
123	T022	J07	4	6250	1553	327	start	M07
123	T023	J08	4	6250	1553	62	start	M08
139	T003	J03	0	6250	466	45	finish	M02
139	T024	J08	4	6250	1553	91	start	M02
141	T027	J09	4	6250	1553	291	submit	-
170	T028	J10	4	1250	466	178	submit	-
170	T028	J10	4	1250	466	178	start	M05
185	T023	J08	4	6250	1553	62	finish	M08
185	T025	J08	4	6250	1553	62	start	M08
189	T013	J05	0	625	387	89	finish	M04
190	T009	J05	0	625	387	90	finish	M03
190	T019	J05	0	625	387	90	finish	M04
191	T005	J05	0	625	387	91	finish	M03
192	T006	J05	0	625	387	92	finish	M03
192	T010	J05	0	625	387	92	finish	M04
193	T007	J05	0	625	387	93	finish	M03
193	T018	J05	0	625	387	93	finish	M04
198	T011	J05	0	625	387	98	finish	M04
199	T012	J05	0	625	387	99	finish	M04
202	T008	J05	0	625	387	102	finish	M03
208	T020	J05	0	625	387	108	finish	M05
212	T016	J05	0	625	387	112	finish	M04
212	T017	J05	0	625	387	112	finish	M04
214	T029	J11	0	6250	1244	149	submit	-
215	T030	J12	0	6250	1244	165	submit	-
218	T015	J05	0	625	387	118	finish	M04
230	T024	J08	4	6250	1553	91	finish	M02
230	T026	J08	4	6250	1553	132	start	M02
247	T025	J08	4	6250	1553	62	finish	M08
247	T027	J09	4	6250	1553	291	start	M08
274	T031	J13	1	62	398	24	submit	-
274	T031	J13	1	62	398	24	start	M03
275	T032	J14	9	6250	1590	74	submit	-
275	T004	J04	0	3125	795	6827	evict	M03
275	T031	J13	1	62	398	24	evict	M03
275	T032	J14	9	6250	1590	74	start	M03
275	T031	J13	1	62	398	24	start	M04
275	T004	J04	0	3125	795	6827	start	M04
288	T014	J05	0	625	387	188	finish	M04
299	T031	J13	1	62	398	24	finish	M04
334	T033	J15	0	3125	311	330	submit	-
334	T033	J15	0	3125	311	330	start	M04
339	T034	J16	8	3125	388	19	submit	-
339	T034	J16	8	3125	388	19	start	M05
348	T028	J10	4	1250	466	178	finish	M05
349	T032	J14	9	6250	1590	74	finish	M03
349	T029	J11	0	6250	1244	149	start	M03
358	T034	J16	8	3125	388	19	finish	M05

time	task	job	priority	CPU	memory	run time	event	machine
358	T030	J12	0	6250	1244	165	start	M05
362	T026	J08	4	6250	1553	132	finish	M02
394	T035	J17	9	6250	3180	214	submit	-
394	T035	J17	9	6250	3180	214	start	M02
449	T021	J06	8	6250	1553	347	finish	M06
450	T022	J07	4	6250	1553	327	finish	M07
485	T036	J18	4	6250	1553	78	submit	-
485	T037	J18	4	6250	1553	75	submit	-
485	T038	J18	4	6250	1553	83	submit	-
485	T039	J18	4	6250	1553	92	submit	-
485	T036	J18	4	6250	1553	78	start	M06
485	T037	J18	4	6250	1553	75	start	M07
495	T001	J01	4	62	398	447	finish	M01
498	T029	J11	0	6250	1244	149	finish	M03
498	T038	J18	4	6250	1553	83	start	M03
504	T040	J19	4	1250	466	180	submit	-
504	T040	J19	4	1250	466	180	start	M01
505	T041	J20	4	1250	466	178	submit	-
505	T042	J20	4	1250	466	177	submit	-
505	T043	J20	4	1250	466	179	submit	-
505	T044	J20	4	1250	466	177	submit	-
523	T030	J12	0	6250	1244	165	finish	M05
523	T039	J18	4	6250	1553	92	start	M05
538	T027	J09	4	6250	1553	291	finish	M08
538	T041	J20	4	1250	466	178	start	M08
538	T042	J20	4	1250	466	177	start	M08
538	T043	J20	4	1250	466	179	start	M08
538	T044	J20	4	1250	466	177	start	M08
541	T037	J18	4	6250	1553	75	fail	M07
563	T036	J18	4	6250	1553	78	finish	M06
563	T037	J18	4	6250	1553	75	start	M06
581	T038	J18	4	6250	1553	83	finish	M03
608	T035	J17	9	6250	3180	214	finish	M02
615	T039	J18	4	6250	1553	92	finish	M05
638	T037	J18	4	6250	1553	75	finish	M06
664	T033	J15	0	3125	311	330	finish	M04
684	T040	J19	4	1250	466	180	finish	M01
684	T045	J21	8	2499	3339	1159	submit	-
684	T045	J21	8	2499	3339	1159	start	M02
694	T046	J22	0	6250	466	32	submit	-
694	T046	J22	0	6250	466	32	start	M03
697	T047	J23	9	6250	3180	214	submit	-
697	T048	J24	9	4688	6360	521	submit	-
697	T049	J25	1	1250	1590	309	submit	-
697	T046	J22	0	6250	466	32	evict	M03
697	T047	J23	9	6250	3180	214	start	M03
697	T049	J25	1	1250	1590	309	start	M02
697	T046	J22	0	6250	466	32	start	M05
715	T042	J20	4	1250	466	177	finish	M08
715	T044	J20	4	1250	466	177	finish	M08
716	T041	J20	4	1250	466	178	finish	M08
717	T043	J20	4	1250	466	179	finish	M08
720	T050	J26	4	2499	3339	177	submit	-
720	T051	J26	4	2499	3339	227	submit	-
720	T052	J26	4	2499	3339	186	submit	-
720	T053	J26	4	2499	3339	249	submit	-
720	T054	J27	4	3125	1125	30	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
720	T050	J26	4	2499	3339	177	start	M02
720	T051	J26	4	2499	3339	227	start	M04
720	T052	J26	4	2499	3339	186	start	M08
720	T054	J27	4	3125	1125	30	start	M06
729	T046	J22	0	6250	466	32	finish	M05
750	T054	J27	4	3125	1125	30	finish	M06
757	T055	J28	0	6250	1244	468	submit	-
757	T055	J28	0	6250	1244	468	start	M05
758	T056	J29	0	6250	1244	120	submit	-
758	T056	J29	0	6250	1244	120	start	M06
759	T057	J30	0	6250	1244	144	submit	-
819	T058	J31	4	2499	3339	308	submit	-
819	T059	J31	4	2499	3339	308	submit	-
819	T060	J31	4	2499	3339	335	submit	-
819	T061	J31	4	2499	3339	290	submit	-
878	T056	J29	0	6250	1244	120	finish	M06
878	T057	J30	0	6250	1244	144	start	M06
897	T050	J26	4	2499	3339	177	finish	M02
897	T053	J26	4	2499	3339	249	start	M02
906	T052	J26	4	2499	3339	186	finish	M08
906	T058	J31	4	2499	3339	308	start	M08
911	T047	J23	9	6250	3180	214	finish	M03
911	T062	J32	4	1250	466	232	submit	-
911	T059	J31	4	2499	3339	308	start	M03
911	T062	J32	4	1250	466	232	start	M01
913	T063	J33	4	1250	466	219	submit	-
913	T064	J33	4	1250	466	191	submit	-
913	T065	J33	4	1250	466	199	submit	-
913	T066	J33	4	1250	466	194	submit	-
913	T063	J33	4	1250	466	219	start	M03
913	T064	J33	4	1250	466	191	start	M03
913	T065	J33	4	1250	466	199	start	M03
913	T066	J33	4	1250	466	194	start	M08
934	T067	J34	0	6250	1244	223	submit	-
935	T068	J35	0	6250	1244	228	submit	-
947	T051	J26	4	2499	3339	227	finish	M04
947	T060	J31	4	2499	3339	335	start	M04
952	T069	J36	4	625	155	25	submit	-
952	T070	J37	8	625	1553	782	submit	-
952	T070	J37	8	625	1553	782	start	M04
952	T069	J36	4	625	155	25	start	M01
977	T069	J36	4	625	155	25	finish	M01
993	T071	J38	0	1250	16	19	submit	-
993	T071	J38	0	1250	16	19	start	M01
1006	T049	J25	1	1250	1590	309	finish	M02
1012	T071	J38	0	1250	16	19	finish	M01
1015	T072	J39	4	1566	204	330	submit	-
1015	T073	J39	4	1566	204	328	submit	-
1015	T074	J39	4	1566	204	327	submit	-
1015	T075	J39	4	1566	204	329	submit	-
1015	T072	J39	4	1566	204	330	start	M01
1015	T073	J39	4	1566	204	328	start	M08
1018	T076	J40	4	6250	622	514	submit	-
1018	T077	J40	4	6250	622	528	submit	-
1018	T078	J40	4	6250	622	508	submit	-
1018	T079	J40	4	6250	622	421	submit	-
1022	T057	J30	0	6250	1244	144	finish	M06

time	task	job	priority	CPU	memory	run time	event	machine
1022	T074	J39	4	1566	204	327	start	M06
1022	T075	J39	4	1566	204	329	start	M06
1061	T080	J41	1	937	466	308	submit	-
1061	T080	J41	1	937	466	308	start	M02
1063	T081	J42	1	1250	610	306	submit	-
1063	T082	J42	1	1250	610	305	submit	-
1063	T083	J42	1	1250	610	303	submit	-
1063	T081	J42	1	1250	610	306	start	M06
1063	T082	J42	1	1250	610	305	start	M06
1066	T084	J43	1	937	549	138	submit	-
1066	T085	J43	1	937	549	81	submit	-
1066	T086	J43	1	937	549	88	submit	-
1066	T087	J43	1	937	549	78	submit	-
1066	T088	J43	1	937	549	68	submit	-
1066	T089	J43	1	937	549	196	submit	-
1066	T090	J43	1	937	549	76	submit	-
1066	T091	J43	1	937	549	76	submit	-
1066	T092	J43	1	937	549	88	submit	-
1066	T093	J43	1	937	549	106	submit	-
1066	T094	J43	1	937	549	88	submit	-
1066	T095	J43	1	937	549	90	submit	-
1066	T096	J43	1	937	549	77	submit	-
1066	T097	J43	1	937	549	81	submit	-
1066	T098	J43	1	937	549	111	submit	-
1066	T099	J43	1	937	549	72	submit	-
1104	T064	J33	4	1250	466	191	finish	M03
1104	T083	J42	1	1250	610	303	start	M03
1107	T066	J33	4	1250	466	194	finish	M08
1107	T084	J43	1	937	549	138	start	M08
1107	T085	J43	1	937	549	81	start	M08
1112	T065	J33	4	1250	466	199	finish	M03
1112	T086	J43	1	937	549	88	start	M03
1124	T100	J44	1	1874	777	432	submit	-
1132	T063	J33	4	1250	466	219	finish	M03
1132	T087	J43	1	937	549	78	start	M03
1137	T101	J45	8	6250	3180	918	submit	-
1137	T102	J46	8	625	466	337	submit	-
1137	T102	J46	8	625	466	337	start	M03
1139	T103	J47	4	6250	2704	713	submit	-
1139	T104	J47	4	6250	2704	724	submit	-
1139	T105	J47	4	6250	2704	725	submit	-
1139	T106	J47	4	6250	2704	717	submit	-
1143	T062	J32	4	1250	466	232	finish	M01
1143	T088	J43	1	937	549	68	start	M01
1146	T053	J26	4	2499	3339	249	finish	M02
1146	T061	J31	4	2499	3339	290	start	M02
1154	T107	J48	8	2499	3339	1099	submit	-
1155	T108	J49	4	6250	1553	62	submit	-
1155	T109	J49	4	6250	1553	145	submit	-
1155	T110	J49	4	6250	1553	64	submit	-
1155	T111	J49	4	6250	1553	64	submit	-
1183	T112	J50	8	937	777	275	submit	-
1188	T085	J43	1	937	549	81	finish	M08
1188	T112	J50	8	937	777	275	start	M08
1195	T113	J51	1	1250	1166	341	submit	-
1199	T114	J52	0	3125	777	91	submit	-
1199	T115	J52	0	3125	777	96	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
1199	T116	J52	0	3125	777	117	submit	-
1199	T117	J52	0	3125	777	101	submit	-
1199	T118	J52	0	3125	777	91	submit	-
1199	T119	J52	0	3125	777	104	submit	-
1199	T120	J52	0	3125	777	81	submit	-
1199	T121	J52	0	3125	777	92	submit	-
1199	T122	J52	0	3125	777	104	submit	-
1199	T123	J52	0	3125	777	167	submit	-
1200	T086	J43	1	937	549	88	finish	M03
1200	T089	J43	1	937	549	196	start	M03

Cell window 03: Events

Time 275: task T032 (priority 9) evicted task T004 (priority 0) from machine M03.

Time 275: task T032 (priority 9) evicted task T031 (priority 1) from machine M03.

Time 275: task T004 restarted on machine M04.

Time 275: task T031 restarted on machine M04.

Time 541: machine M07 failed; task T037 stopped and went back to the queue.

Time 563: task T037 restarted on machine M06.

Time 697: task T047 (priority 9) evicted task T046 (priority 0) from machine M03.

Time 697: task T046 restarted on machine M05.

Longest waits (submission to first start, tasks started by the window end): task T061 waited 327 s (submitted at 819, first started at 1146); task T053 waited 177 s (submitted at 720, first started at 897); task T030 waited 143 s (submitted at 215, first started at 358); task T029 waited 135 s (submitted at 214, first started at 349); task T089 waited 134 s (submitted at 1066, first started at 1200).

Cell window 03: Queue at window end

State after the events at time 1200.

Tasks running at time 1200 (remaining run time = seconds still to run after time 1200):

task	job	priority	CPU	memory	machine	started	remaining
T002	J02	1	3125	795	M01	94	5725
T072	J39	4	1566	204	M01	1015	145
T088	J43	1	937	549	M01	1143	11
T045	J21	8	2499	3339	M02	684	643
T061	J31	4	2499	3339	M02	1146	236
T080	J41	1	937	466	M02	1061	169
T059	J31	4	2499	3339	M03	911	19
T083	J42	1	1250	610	M03	1104	207
T087	J43	1	937	549	M03	1132	10
T089	J43	1	937	549	M03	1200	196
T102	J46	8	625	466	M03	1137	274
T004	J04	0	3125	795	M04	275	5902
T060	J31	4	2499	3339	M04	947	82
T070	J37	8	625	1553	M04	952	534
T055	J28	0	6250	1244	M05	757	25
T074	J39	4	1566	204	M06	1022	149
T075	J39	4	1566	204	M06	1022	151
T081	J42	1	1250	610	M06	1063	169
T082	J42	1	1250	610	M06	1063	168
T058	J31	4	2499	3339	M08	906	14

task	job	priority	CPU	memory	machine	started	remaining
T073	J39	4	1566	204	M08	1015	143
T084	J43	1	937	549	M08	1107	45
T112	J50	8	937	777	M08	1188	263

Tasks waiting at time 1200, in queue order (each will run its full run time once started):

task	job	priority	CPU	memory	submitted	run time
T048	J24	9	4688	6360	697	521
T101	J45	8	6250	3180	1137	918
T107	J48	8	2499	3339	1154	1099
T076	J40	4	6250	622	1018	514
T077	J40	4	6250	622	1018	528
T078	J40	4	6250	622	1018	508
T079	J40	4	6250	622	1018	421
T103	J47	4	6250	2704	1139	713
T104	J47	4	6250	2704	1139	724
T105	J47	4	6250	2704	1139	725
T106	J47	4	6250	2704	1139	717
T108	J49	4	6250	1553	1155	62
T109	J49	4	6250	1553	1155	145
T110	J49	4	6250	1553	1155	64
T111	J49	4	6250	1553	1155	64
T090	J43	1	937	549	1066	76
T091	J43	1	937	549	1066	76
T092	J43	1	937	549	1066	88
T093	J43	1	937	549	1066	106
T094	J43	1	937	549	1066	88
T095	J43	1	937	549	1066	90
T096	J43	1	937	549	1066	77
T097	J43	1	937	549	1066	81
T098	J43	1	937	549	1066	111
T099	J43	1	937	549	1066	72
T100	J44	1	1874	777	1124	432
T113	J51	1	1250	1166	1195	341
T067	J34	0	6250	1244	934	223
T068	J35	0	6250	1244	935	228
T114	J52	0	3125	777	1199	91
T115	J52	0	3125	777	1199	96
T116	J52	0	3125	777	1199	117
T117	J52	0	3125	777	1199	101
T118	J52	0	3125	777	1199	91
T119	J52	0	3125	777	1199	104
T120	J52	0	3125	777	1199	81
T121	J52	0	3125	777	1199	92
T122	J52	0	3125	777	1199	104
T123	J52	0	3125	777	1199	167

Machine M07 is down at time 1200; it returns at time 1505.

Cell window 20

Times 0 to 1200. The cluster has 8 machines. 111 tasks of 47 jobs are submitted during the window.

Cell window 20: Machines

machine	CPU	memory	fails at	returns at
M01	6250	3116	-	-
M02	6250	3116	-	-
M03	6250	6244	-	-
M04	6250	6244	-	-
M05	6250	9363	574	never
M06	6250	3116	-	-
M07	6250	6244	-	-
M08	6250	3116	-	-

Cell window 20: Task log

time	task	job	priority	CPU	memory	run time	event	machine
83	T001	J01	6	937	197	127	submit	-
83	T002	J01	6	937	197	123	submit	-
83	T003	J01	6	937	197	111	submit	-
83	T004	J01	6	937	197	113	submit	-
83	T001	J01	6	937	197	127	start	M01
83	T002	J01	6	937	197	123	start	M01
83	T003	J01	6	937	197	111	start	M01
83	T004	J01	6	937	197	113	start	M01
97	T005	J02	6	625	280	52	submit	-
97	T005	J02	6	625	280	52	start	M01
130	T006	J03	0	6250	466	39	submit	-
130	T006	J03	0	6250	466	39	start	M02
132	T007	J04	9	4688	6360	528	submit	-
132	T007	J04	9	4688	6360	528	start	M05
134	T008	J05	8	625	398	305	submit	-
134	T008	J05	8	625	398	305	start	M01
149	T005	J02	6	625	280	52	finish	M01
149	T009	J06	4	1250	466	252	submit	-
149	T009	J06	4	1250	466	252	start	M01
167	T010	J07	1	1250	466	176	submit	-
167	T010	J07	1	1250	466	176	start	M03
169	T006	J03	0	6250	466	39	finish	M02
178	T011	J08	0	937	1193	1092	submit	-
178	T012	J08	0	937	1193	1093	submit	-
178	T013	J08	0	937	1193	1094	submit	-
178	T014	J08	0	937	1193	1093	submit	-
178	T015	J08	0	937	1193	1093	submit	-
178	T016	J08	0	937	1193	1093	submit	-
178	T017	J08	0	937	1193	1093	submit	-
178	T018	J08	0	937	1193	1094	submit	-
178	T019	J08	0	937	1193	1094	submit	-
178	T020	J08	0	937	1193	1094	submit	-
178	T021	J08	0	937	1193	1093	submit	-
178	T022	J08	0	937	1193	1094	submit	-
178	T023	J08	0	937	1193	1093	submit	-
178	T024	J08	0	937	1193	1094	submit	-
178	T025	J08	0	937	1193	1094	submit	-
178	T026	J08	0	937	1193	1092	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
178	T011	J08	0	937	1193	1092	start	M02
178	T012	J08	0	937	1193	1093	start	M02
178	T013	J08	0	937	1193	1094	start	M03
178	T014	J08	0	937	1193	1093	start	M03
178	T015	J08	0	937	1193	1093	start	M03
178	T016	J08	0	937	1193	1093	start	M03
178	T017	J08	0	937	1193	1093	start	M04
178	T018	J08	0	937	1193	1094	start	M04
178	T019	J08	0	937	1193	1094	start	M04
178	T020	J08	0	937	1193	1094	start	M04
178	T021	J08	0	937	1193	1093	start	M04
178	T022	J08	0	937	1193	1094	start	M05
178	T023	J08	0	937	1193	1093	start	M06
178	T024	J08	0	937	1193	1094	start	M06
178	T025	J08	0	937	1193	1094	start	M07
178	T026	J08	0	937	1193	1092	start	M07
192	T027	J09	0	6250	1244	422	submit	-
192	T027	J09	0	6250	1244	422	start	M08
194	T003	J01	6	937	197	111	finish	M01
194	T028	J10	0	6250	1244	87	submit	-
196	T004	J01	6	937	197	113	finish	M01
200	T029	J11	4	3125	1125	46	submit	-
200	T029	J11	4	3125	1125	46	start	M07
206	T002	J01	6	937	197	123	finish	M01
210	T001	J01	6	937	197	127	finish	M01
246	T029	J11	4	3125	1125	46	finish	M07
343	T010	J07	1	1250	466	176	finish	M03
346	T030	J12	6	937	249	65	submit	-
346	T031	J12	6	937	249	65	submit	-
346	T032	J12	6	937	249	131	submit	-
346	T033	J12	6	937	249	145	submit	-
346	T030	J12	6	937	249	65	start	M01
346	T031	J12	6	937	249	65	start	M01
346	T032	J12	6	937	249	131	start	M01
346	T033	J12	6	937	249	145	start	M01
350	T034	J13	1	1562	1553	1107	submit	-
350	T034	J13	1	1562	1553	1107	start	M07
386	T035	J14	8	625	311	326	submit	-
386	T035	J14	8	625	311	326	start	M01
395	T036	J15	1	1843	1553	1046	submit	-
395	T036	J15	1	1843	1553	1046	start	M07
396	T037	J16	0	1250	316	1045	submit	-
396	T037	J16	0	1250	316	1045	start	M02
401	T009	J06	4	1250	466	252	finish	M01
411	T030	J12	6	937	249	65	finish	M01
411	T031	J12	6	937	249	65	finish	M01
419	T038	J17	1	3125	2386	966	submit	-
439	T008	J05	8	625	398	305	finish	M01
439	T039	J18	4	625	155	31	submit	-
439	T039	J18	4	625	155	31	start	M01
470	T039	J18	4	625	155	31	finish	M01
477	T032	J12	6	937	249	131	finish	M01
477	T038	J17	1	3125	2386	966	start	M01
491	T033	J12	6	937	249	145	finish	M01
574	T007	J04	9	4688	6360	528	fail	M05
574	T022	J08	0	937	1193	1094	fail	M05
574	T022	J08	0	937	1193	1094	start	M03

time	task	job	priority	CPU	memory	run time	event	machine
577	T040	J19	4	1250	466	231	submit	-
577	T040	J19	4	1250	466	231	start	M06
614	T027	J09	0	6250	1244	422	finish	M08
614	T041	J20	1	937	777	65	submit	-
614	T041	J20	1	937	777	65	start	M08
615	T042	J21	9	1250	1590	53	submit	-
615	T042	J21	9	1250	1590	53	start	M08
623	T043	J22	4	3125	224	71	submit	-
623	T043	J22	4	3125	224	71	start	M02
652	T044	J23	6	625	197	76	submit	-
652	T044	J23	6	625	197	76	start	M01
668	T042	J21	9	1250	1590	53	finish	M08
669	T045	J24	0	3125	388	287	submit	-
670	T046	J25	9	62	466	2775	submit	-
670	T046	J25	9	62	466	2775	start	M07
676	T047	J26	8	625	199	683	submit	-
676	T047	J26	8	625	199	683	start	M01
679	T041	J20	1	937	777	65	finish	M08
679	T028	J10	0	6250	1244	87	start	M08
691	T048	J27	4	3125	468	226	submit	-
691	T049	J27	4	3125	468	241	submit	-
691	T050	J27	4	3125	468	241	submit	-
691	T051	J27	4	3125	468	234	submit	-
694	T043	J22	4	3125	224	71	finish	M02
694	T045	J24	0	3125	388	287	start	M02
700	T052	J28	9	1268	466	2728	submit	-
700	T038	J17	1	3125	2386	966	evict	M01
700	T052	J28	9	1268	466	2728	start	M01
702	T053	J29	1	687	932	2723	submit	-
702	T054	J29	1	687	932	2723	submit	-
703	T055	J30	0	687	932	2637	submit	-
703	T056	J30	0	687	932	2612	submit	-
703	T057	J30	0	687	932	2620	submit	-
703	T058	J30	0	687	932	2614	submit	-
703	T059	J30	0	687	932	2619	submit	-
703	T060	J30	0	687	932	2617	submit	-
703	T061	J30	0	687	932	2624	submit	-
703	T062	J30	0	687	932	2605	submit	-
703	T063	J30	0	687	932	2624	submit	-
703	T064	J30	0	687	932	2613	submit	-
703	T065	J30	0	687	932	2713	submit	-
703	T066	J30	0	687	932	2613	submit	-
703	T067	J30	0	687	932	2620	submit	-
703	T068	J30	0	687	932	2604	submit	-
703	T069	J30	0	687	932	2624	submit	-
703	T070	J30	0	687	932	2625	submit	-
712	T035	J14	8	625	311	326	finish	M01
712	T048	J27	4	3125	468	226	start	M01
728	T044	J23	6	625	197	76	finish	M01
728	T053	J29	1	687	932	2723	start	M01
732	T071	J31	9	6250	3180	236	submit	-
732	T022	J08	0	937	1193	1094	evict	M03
732	T016	J08	0	937	1193	1093	evict	M03
732	T015	J08	0	937	1193	1093	evict	M03
732	T014	J08	0	937	1193	1093	evict	M03
732	T013	J08	0	937	1193	1094	evict	M03
732	T071	J31	9	6250	3180	236	start	M03

time	task	job	priority	CPU	memory	run time	event	machine
741	T072	J32	4	625	311	23	submit	-
753	T073	J33	8	625	398	326	submit	-
766	T028	J10	0	6250	1244	87	finish	M08
766	T074	J34	1	937	1590	1173	submit	-
766	T073	J33	8	625	398	326	start	M08
766	T049	J27	4	3125	468	241	start	M08
766	T072	J32	4	625	311	23	start	M08
766	T054	J29	1	687	932	2723	start	M08
766	T055	J30	0	687	932	2637	start	M08
768	T075	J35	4	1250	466	324	submit	-
768	T076	J35	4	1250	466	311	submit	-
768	T077	J35	4	1250	466	311	submit	-
768	T078	J35	4	1250	466	303	submit	-
768	T079	J36	0	1250	919	1171	submit	-
770	T080	J37	0	937	244	81	submit	-
770	T081	J37	0	937	244	72	submit	-
770	T082	J37	0	937	244	77	submit	-
770	T083	J37	0	937	244	77	submit	-
770	T084	J37	0	937	244	82	submit	-
770	T085	J37	0	937	244	72	submit	-
770	T086	J37	0	937	244	90	submit	-
770	T087	J37	0	937	244	72	submit	-
770	T088	J37	0	937	244	77	submit	-
770	T089	J37	0	937	244	81	submit	-
770	T090	J37	0	937	244	77	submit	-
770	T091	J37	0	937	244	78	submit	-
770	T092	J37	0	937	244	70	submit	-
770	T093	J37	0	937	244	108	submit	-
770	T094	J37	0	937	244	89	submit	-
770	T095	J37	0	937	244	73	submit	-
770	T080	J37	0	937	244	81	start	M04
770	T081	J37	0	937	244	72	start	M06
789	T072	J32	4	625	311	23	finish	M08
789	T082	J37	0	937	244	77	start	M08
805	T096	J38	8	5627	3180	261	submit	-
808	T040	J19	4	1250	466	231	finish	M06
808	T050	J27	4	3125	468	241	start	M06
842	T081	J37	0	937	244	72	finish	M06
842	T083	J37	0	937	244	77	start	M06
850	T097	J39	0	6250	1244	58	submit	-
851	T080	J37	0	937	244	81	finish	M04
851	T084	J37	0	937	244	82	start	M04
866	T082	J37	0	937	244	77	finish	M08
866	T085	J37	0	937	244	72	start	M08
919	T083	J37	0	937	244	77	finish	M06
919	T086	J37	0	937	244	90	start	M06
932	T098	J40	4	1250	155	12	submit	-
933	T084	J37	0	937	244	82	finish	M04
933	T098	J40	4	1250	155	12	start	M04
938	T048	J27	4	3125	468	226	finish	M01
938	T085	J37	0	937	244	72	finish	M08
938	T051	J27	4	3125	468	234	start	M01
938	T087	J37	0	937	244	72	start	M08
945	T098	J40	4	1250	155	12	finish	M04
945	T088	J37	0	937	244	77	start	M04
968	T071	J31	9	6250	3180	236	finish	M03
968	T096	J38	8	5627	3180	261	start	M03

time	task	job	priority	CPU	memory	run time	event	machine
981	T045	J24	0	3125	388	287	finish	M02
981	T089	J37	0	937	244	81	start	M02
1007	T049	J27	4	3125	468	241	finish	M08
1007	T075	J35	4	1250	466	324	start	M08
1009	T086	J37	0	937	244	90	finish	M06
1009	T090	J37	0	937	244	77	start	M06
1010	T087	J37	0	937	244	72	finish	M08
1010	T091	J37	0	937	244	78	start	M08
1022	T088	J37	0	937	244	77	finish	M04
1022	T092	J37	0	937	244	70	start	M04
1023	T099	J41	4	1874	1540	456	submit	-
1023	T100	J41	4	1874	1540	446	submit	-
1023	T101	J41	4	1874	1540	449	submit	-
1023	T102	J41	4	1874	1540	465	submit	-
1031	T103	J42	9	6250	233	114	submit	-
1031	T089	J37	0	937	244	81	evict	M02
1031	T037	J16	0	1250	316	1045	evict	M02
1031	T012	J08	0	937	1193	1093	evict	M02
1031	T011	J08	0	937	1193	1092	evict	M02
1031	T103	J42	9	6250	233	114	start	M02
1047	T104	J43	4	1250	398	32	submit	-
1049	T050	J27	4	3125	468	241	finish	M06
1049	T076	J35	4	1250	466	311	start	M06
1061	T105	J44	8	6250	1166	243	submit	-
1085	T106	J45	8	625	398	518	submit	-
1086	T090	J37	0	937	244	77	finish	M06
1086	T089	J37	0	937	244	81	start	M06
1088	T091	J37	0	937	244	78	finish	M08
1088	T037	J16	0	1250	316	1045	start	M08
1092	T073	J33	8	625	398	326	finish	M08
1092	T092	J37	0	937	244	70	finish	M04
1092	T106	J45	8	625	398	518	start	M08
1092	T093	J37	0	937	244	108	start	M04
1145	T103	J42	9	6250	233	114	finish	M02
1145	T105	J44	8	6250	1166	243	start	M02
1162	T107	J46	8	5627	1590	268	submit	-
1167	T089	J37	0	937	244	81	finish	M06
1167	T094	J37	0	937	244	89	start	M06
1172	T051	J27	4	3125	468	234	finish	M01
1172	T077	J35	4	1250	466	311	start	M01
1172	T078	J35	4	1250	466	303	start	M01
1172	T095	J37	0	937	244	73	start	M01
1188	T108	J47	4	3125	777	151	submit	-
1188	T109	J47	4	3125	777	153	submit	-
1188	T110	J47	4	3125	777	151	submit	-
1188	T111	J47	4	3125	777	151	submit	-
1200	T093	J37	0	937	244	108	finish	M04

Cell window 20: Events

Time 574: machine M05 failed; tasks T007, T022 stopped and went back to the queue.

Time 574: task T022 restarted on machine M03.

Time 700: task T052 (priority 9) evicted task T038 (priority 1) from machine M01.

Time 732: task T071 (priority 9) evicted task T013 (priority 0) from machine M03.

Time 732: task T071 (priority 9) evicted task T014 (priority 0) from machine M03.

Time 732: task T071 (priority 9) evicted task T015 (priority 0) from machine M03.

Time 732: task T071 (priority 9) evicted task T016 (priority 0) from machine M03.

Time 732: task T071 (priority 9) evicted task T022 (priority 0) from machine M03.

Time 1031: task T103 (priority 9) evicted task T011 (priority 0) from machine M02.

Time 1031: task T103 (priority 9) evicted task T012 (priority 0) from machine M02.

Time 1031: task T103 (priority 9) evicted task T037 (priority 0) from machine M02.

Time 1031: task T103 (priority 9) evicted task T089 (priority 0) from machine M02.

Time 1086: task T089 restarted on machine M06.

Time 1088: task T037 restarted on machine M08.

Longest waits (submission to first start, tasks started by the window end): task T028 waited 485 s (submitted at 194, first started at 679); task T077 waited 404 s (submitted at 768, first started at 1172); task T078 waited 404 s (submitted at 768, first started at 1172); task T095 waited 402 s (submitted at 770, first started at 1172); task T094 waited 397 s (submitted at 770, first started at 1167).

Cell window 20: Queue at window end

State after the events at time 1200.

Tasks running at time 1200 (remaining run time = seconds still to run after time 1200):

task	job	priority	CPU	memory	machine	started	remaining
T047	J26	8	625	199	M01	676	159
T052	J28	9	1268	466	M01	700	2228
T053	J29	1	687	932	M01	728	2251
T077	J35	4	1250	466	M01	1172	283
T078	J35	4	1250	466	M01	1172	275
T095	J37	0	937	244	M01	1172	45
T105	J44	8	6250	1166	M02	1145	188
T096	J38	8	5627	3180	M03	968	29
T017	J08	0	937	1193	M04	178	71
T018	J08	0	937	1193	M04	178	72
T019	J08	0	937	1193	M04	178	72
T020	J08	0	937	1193	M04	178	72
T021	J08	0	937	1193	M04	178	71
T023	J08	0	937	1193	M06	178	71
T024	J08	0	937	1193	M06	178	72
T076	J35	4	1250	466	M06	1049	160
T094	J37	0	937	244	M06	1167	56
T025	J08	0	937	1193	M07	178	72
T026	J08	0	937	1193	M07	178	70
T034	J13	1	1562	1553	M07	350	257
T036	J15	1	1843	1553	M07	395	241
T046	J25	9	62	466	M07	670	2245
T037	J16	0	1250	316	M08	1088	933
T054	J29	1	687	932	M08	766	2289
T055	J30	0	687	932	M08	766	2203
T075	J35	4	1250	466	M08	1007	131
T106	J45	8	625	398	M08	1092	410

Tasks waiting at time 1200, in queue order (each will run its full run time once started):

task	job	priority	CPU	memory	submitted	run time
T007	J04	9	4688	6360	132	528
T107	J46	8	5627	1590	1162	268
T099	J41	4	1874	1540	1023	456

task	job	priority	CPU	memory	submitted	run time
T100	J41	4	1874	1540	1023	446
T101	J41	4	1874	1540	1023	449
T102	J41	4	1874	1540	1023	465
T104	J43	4	1250	398	1047	32
T108	J47	4	3125	777	1188	151
T109	J47	4	3125	777	1188	153
T110	J47	4	3125	777	1188	151
T111	J47	4	3125	777	1188	151
T038	J17	1	3125	2386	419	966
T074	J34	1	937	1590	766	1173
T011	J08	0	937	1193	178	1092
T012	J08	0	937	1193	178	1093
T013	J08	0	937	1193	178	1094
T014	J08	0	937	1193	178	1093
T015	J08	0	937	1193	178	1093
T016	J08	0	937	1193	178	1093
T022	J08	0	937	1193	178	1094
T056	J30	0	687	932	703	2612
T057	J30	0	687	932	703	2620
T058	J30	0	687	932	703	2614
T059	J30	0	687	932	703	2619
T060	J30	0	687	932	703	2617
T061	J30	0	687	932	703	2624
T062	J30	0	687	932	703	2605
T063	J30	0	687	932	703	2624
T064	J30	0	687	932	703	2613
T065	J30	0	687	932	703	2713
T066	J30	0	687	932	703	2613
T067	J30	0	687	932	703	2620
T068	J30	0	687	932	703	2604
T069	J30	0	687	932	703	2624
T070	J30	0	687	932	703	2625
T079	J36	0	1250	919	768	1171
T097	J39	0	6250	1244	850	58

Machine M05 is down at time 1200; it does not return.

Cell window 23

Times 0 to 1200. The cluster has 8 machines. 207 tasks of 94 jobs are submitted during the window.

Cell window 23: Machines

machine	CPU	memory	fails at	returns at
M01	6250	3116	-	-
M02	6250	6244	-	-
M03	6250	6244	-	-
M04	6250	6244	416	824
M05	6250	6244	-	-
M06	6250	6244	-	-
M07	6250	3116	-	-
M08	6250	3116	-	-

Cell window 23: Task log

time	task	job	priority	CPU	memory	run time	event	machine
1	T001	J01	4	1250	466	191	submit	-
1	T002	J01	4	1250	466	175	submit	-
1	T003	J01	4	1250	466	190	submit	-
1	T004	J01	4	1250	466	190	submit	-
1	T001	J01	4	1250	466	191	start	M01
1	T002	J01	4	1250	466	175	start	M01
1	T003	J01	4	1250	466	190	start	M01
1	T004	J01	4	1250	466	190	start	M01
97	T005	J02	0	6250	1244	53	submit	-
97	T006	J03	0	6250	466	44	submit	-
97	T007	J04	0	6250	1244	65	submit	-
97	T005	J02	0	6250	1244	53	start	M02
97	T006	J03	0	6250	466	44	start	M03
97	T007	J04	0	6250	1244	65	start	M04
100	T008	J05	9	6250	1590	71	submit	-
100	T008	J05	9	6250	1590	71	start	M05
101	T009	J06	1	1250	1590	393	submit	-
101	T009	J06	1	1250	1590	393	start	M06
116	T010	J07	4	625	155	26	submit	-
116	T011	J08	6	625	280	120	submit	-
116	T012	J09	4	625	155	33	submit	-
116	T013	J10	4	3125	1125	25	submit	-
116	T014	J11	4	625	311	230	submit	-
116	T011	J08	6	625	280	120	start	M01
116	T010	J07	4	625	155	26	start	M01
116	T012	J09	4	625	155	33	start	M06
116	T013	J10	4	3125	1125	25	start	M06
116	T014	J11	4	625	311	230	start	M06
133	T015	J12	6	937	1244	224	submit	-
133	T015	J12	6	937	1244	224	start	M07
134	T016	J13	6	937	229	62	submit	-
134	T017	J13	6	937	229	69	submit	-
134	T018	J13	6	937	229	70	submit	-
134	T019	J13	6	937	229	68	submit	-
134	T016	J13	6	937	229	62	start	M07
134	T017	J13	6	937	229	69	start	M07
134	T018	J13	6	937	229	70	start	M07
134	T019	J13	6	937	229	68	start	M07

time	task	job	priority	CPU	memory	run time	event	machine
141	T006	J03	0	6250	466	44	finish	M03
141	T013	J10	4	3125	1125	25	finish	M06
142	T010	J07	4	625	155	26	finish	M01
147	T020	J14	6	937	1244	76	submit	-
147	T020	J14	6	937	1244	76	start	M03
148	T021	J15	1	1250	466	168	submit	-
148	T021	J15	1	1250	466	168	start	M03
149	T012	J09	4	625	155	33	finish	M06
149	T022	J16	6	645	197	75	submit	-
149	T023	J17	6	937	197	75	submit	-
149	T024	J17	6	937	197	63	submit	-
149	T025	J17	6	937	197	64	submit	-
149	T026	J17	6	937	197	60	submit	-
149	T022	J16	6	645	197	75	start	M03
149	T023	J17	6	937	197	75	start	M03
149	T024	J17	6	937	197	63	start	M03
149	T025	J17	6	937	197	64	start	M03
149	T026	J17	6	937	197	60	start	M06
150	T005	J02	0	6250	1244	53	finish	M02
150	T027	J18	0	625	229	166	submit	-
150	T028	J18	0	625	229	156	submit	-
150	T029	J18	0	625	229	166	submit	-
150	T030	J18	0	625	229	155	submit	-
150	T031	J18	0	625	229	156	submit	-
150	T032	J18	0	625	229	156	submit	-
150	T033	J18	0	625	229	156	submit	-
150	T034	J18	0	625	229	156	submit	-
150	T035	J18	0	625	229	157	submit	-
150	T036	J18	0	625	229	154	submit	-
150	T037	J18	0	625	229	155	submit	-
150	T038	J18	0	625	229	156	submit	-
150	T039	J18	0	625	229	155	submit	-
150	T040	J18	0	625	229	166	submit	-
150	T041	J18	0	625	229	155	submit	-
150	T042	J18	0	625	229	154	submit	-
150	T027	J18	0	625	229	166	start	M01
150	T028	J18	0	625	229	156	start	M02
150	T029	J18	0	625	229	166	start	M02
150	T030	J18	0	625	229	155	start	M02
150	T031	J18	0	625	229	156	start	M02
150	T032	J18	0	625	229	156	start	M02
150	T033	J18	0	625	229	156	start	M02
150	T034	J18	0	625	229	156	start	M02
150	T035	J18	0	625	229	157	start	M02
150	T036	J18	0	625	229	154	start	M02
150	T037	J18	0	625	229	155	start	M02
150	T038	J18	0	625	229	156	start	M06
150	T039	J18	0	625	229	155	start	M06
150	T040	J18	0	625	229	166	start	M06
150	T041	J18	0	625	229	155	start	M06
150	T042	J18	0	625	229	154	start	M06
162	T007	J04	0	6250	1244	65	finish	M04
164	T043	J19	6	6250	622	113	submit	-
164	T043	J19	6	6250	622	113	start	M04
171	T008	J05	9	6250	1590	71	finish	M05
176	T002	J01	4	1250	466	175	finish	M01
191	T003	J01	4	1250	466	190	finish	M01

time	task	job	priority	CPU	memory	run time	event	machine
191	T004	J01	4	1250	466	190	finish	M01
192	T001	J01	4	1250	466	191	finish	M01
196	T016	J13	6	937	229	62	finish	M07
202	T019	J13	6	937	229	68	finish	M07
203	T017	J13	6	937	229	69	finish	M07
204	T018	J13	6	937	229	70	finish	M07
209	T026	J17	6	937	197	60	finish	M06
212	T024	J17	6	937	197	63	finish	M03
213	T025	J17	6	937	197	64	finish	M03
218	T044	J20	6	625	187	23	submit	-
218	T044	J20	6	625	187	23	start	M01
223	T020	J14	6	937	1244	76	finish	M03
224	T022	J16	6	645	197	75	finish	M03
224	T023	J17	6	937	197	75	finish	M03
236	T011	J08	6	625	280	120	finish	M01
241	T044	J20	6	625	187	23	finish	M01
265	T045	J21	6	625	3180	24	submit	-
265	T045	J21	6	625	3180	24	start	M03
277	T043	J19	6	6250	622	113	finish	M04
289	T045	J21	6	625	3180	24	finish	M03
297	T046	J22	6	625	280	358	submit	-
297	T046	J22	6	625	280	358	start	M01
304	T036	J18	0	625	229	154	finish	M02
304	T042	J18	0	625	229	154	finish	M06
305	T030	J18	0	625	229	155	finish	M02
305	T037	J18	0	625	229	155	finish	M02
305	T039	J18	0	625	229	155	finish	M06
305	T041	J18	0	625	229	155	finish	M06
306	T028	J18	0	625	229	156	finish	M02
306	T031	J18	0	625	229	156	finish	M02
306	T032	J18	0	625	229	156	finish	M02
306	T033	J18	0	625	229	156	finish	M02
306	T034	J18	0	625	229	156	finish	M02
306	T038	J18	0	625	229	156	finish	M06
307	T035	J18	0	625	229	157	finish	M02
316	T021	J15	1	1250	466	168	finish	M03
316	T027	J18	0	625	229	166	finish	M01
316	T029	J18	0	625	229	166	finish	M02
316	T040	J18	0	625	229	166	finish	M06
316	T047	J23	1	1250	777	1057	submit	-
316	T047	J23	1	1250	777	1057	start	M01
335	T048	J24	0	3125	466	310	submit	-
335	T048	J24	0	3125	466	310	start	M01
346	T014	J11	4	625	311	230	finish	M06
349	T049	J25	6	937	1244	298	submit	-
349	T050	J26	4	3125	1125	27	submit	-
349	T049	J25	6	937	1244	298	start	M01
349	T050	J26	4	3125	1125	27	start	M02
350	T051	J27	6	937	239	61	submit	-
350	T052	J27	6	937	239	68	submit	-
350	T053	J27	6	937	239	141	submit	-
350	T054	J27	6	937	239	92	submit	-
350	T051	J27	6	937	239	61	start	M02
350	T052	J27	6	937	239	68	start	M02
350	T053	J27	6	937	239	141	start	M02
350	T054	J27	6	937	239	92	start	M03
357	T015	J12	6	937	1244	224	finish	M07

time	task	job	priority	CPU	memory	run time	event	machine
366	T055	J28	0	1688	295	996	submit	-
366	T055	J28	0	1688	295	996	start	M03
374	T056	J29	6	625	280	157	submit	-
374	T056	J29	6	625	280	157	start	M03
376	T050	J26	4	3125	1125	27	finish	M02
395	T057	J30	0	1250	16	18	submit	-
395	T057	J30	0	1250	16	18	start	M02
403	T058	J31	6	625	280	37	submit	-
403	T059	J32	6	937	1244	110	submit	-
403	T058	J31	6	625	280	37	start	M02
403	T059	J32	6	937	1244	110	start	M02
411	T051	J27	6	937	239	61	finish	M02
413	T057	J30	0	1250	16	18	finish	M02
418	T052	J27	6	937	239	68	finish	M02
420	T060	J33	6	634	197	94	submit	-
420	T061	J34	6	937	197	56	submit	-
420	T062	J34	6	937	197	48	submit	-
420	T063	J34	6	937	197	49	submit	-
420	T064	J34	6	937	197	75	submit	-
420	T060	J33	6	634	197	94	start	M02
420	T061	J34	6	937	197	56	start	M02
420	T062	J34	6	937	197	48	start	M02
420	T063	J34	6	937	197	49	start	M02
420	T064	J34	6	937	197	75	start	M03
440	T058	J31	6	625	280	37	finish	M02
442	T054	J27	6	937	239	92	finish	M03
444	T065	J35	2	625	31	83	submit	-
444	T066	J36	2	625	31	28	submit	-
444	T065	J35	2	625	31	83	start	M02
444	T066	J36	2	625	31	28	start	M03
456	T067	J37	6	625	187	18	submit	-
456	T068	J38	8	1874	777	3413	submit	-
456	T068	J38	8	1874	777	3413	start	M03
456	T067	J37	6	625	187	18	start	M05
468	T062	J34	6	937	197	48	finish	M02
469	T063	J34	6	937	197	49	finish	M02
472	T066	J36	2	625	31	28	finish	M03
472	T069	J39	6	4376	777	241	submit	-
472	T070	J39	6	4376	777	270	submit	-
472	T071	J39	6	4376	777	261	submit	-
472	T072	J39	6	4376	777	352	submit	-
472	T069	J39	6	4376	777	241	start	M05
472	T070	J39	6	4376	777	270	start	M06
472	T071	J39	6	4376	777	261	start	M07
472	T072	J39	6	4376	777	352	start	M08
474	T067	J37	6	625	187	18	finish	M05
476	T061	J34	6	937	197	56	finish	M02
491	T053	J27	6	937	239	141	finish	M02
494	T009	J06	1	1250	1590	393	finish	M06
495	T064	J34	6	937	197	75	finish	M03
513	T059	J32	6	937	1244	110	finish	M02
514	T060	J33	6	634	197	94	finish	M02
515	T073	J40	4	3448	1460	599	submit	-
515	T074	J40	4	3448	1460	561	submit	-
515	T075	J40	4	3448	1460	521	submit	-
515	T076	J40	4	3448	1460	595	submit	-
515	T073	J40	4	3448	1460	599	start	M02

time	task	job	priority	CPU	memory	run time	event	machine
527	T065	J35	2	625	31	83	finish	M02
530	T077	J41	4	3125	1224	42	submit	-
531	T056	J29	6	625	280	157	finish	M03
538	T078	J42	1	1250	777	1094	submit	-
538	T078	J42	1	1250	777	1094	start	M02
542	T079	J43	6	625	280	229	submit	-
542	T080	J44	6	625	3180	23	submit	-
542	T079	J43	6	625	280	229	start	M02
542	T080	J44	6	625	3180	23	start	M02
565	T080	J44	6	625	3180	23	finish	M02
572	T081	J45	6	937	1244	184	submit	-
572	T081	J45	6	937	1244	184	start	M03
575	T082	J46	0	6250	1244	62	submit	-
576	T083	J47	0	6250	1244	83	submit	-
586	T084	J48	0	1688	295	1034	submit	-
586	T084	J48	0	1688	295	1034	start	M03
587	T085	J49	4	3125	1125	26	submit	-
600	T086	J50	8	6250	1590	3241	submit	-
605	T087	J51	1	937	777	75	submit	-
605	T087	J51	1	937	777	75	start	M05
607	T088	J52	0	1250	777	73	submit	-
607	T089	J52	0	1250	777	73	submit	-
607	T090	J52	0	1250	777	71	submit	-
607	T088	J52	0	1250	777	73	start	M06
607	T089	J52	0	1250	777	73	start	M07
607	T090	J52	0	1250	777	71	start	M08
608	T091	J53	0	937	277	54	submit	-
608	T092	J53	0	937	277	53	submit	-
608	T093	J53	0	937	277	54	submit	-
608	T094	J53	0	937	277	53	submit	-
608	T095	J53	0	937	277	64	submit	-
608	T096	J53	0	937	277	63	submit	-
608	T097	J53	0	937	277	67	submit	-
608	T098	J53	0	937	277	53	submit	-
608	T099	J53	0	937	277	63	submit	-
608	T100	J53	0	937	277	51	submit	-
608	T101	J53	0	937	277	54	submit	-
608	T102	J53	0	937	277	57	submit	-
608	T103	J53	0	937	277	55	submit	-
608	T104	J53	0	937	277	55	submit	-
608	T105	J53	0	937	277	51	submit	-
608	T106	J53	0	937	277	51	submit	-
608	T091	J53	0	937	277	54	start	M05
614	T107	J54	4	3125	3180	1894	submit	-
614	T108	J54	4	3125	3180	1800	submit	-
614	T109	J54	4	3125	3180	1761	submit	-
614	T110	J54	4	3125	3180	2562	submit	-
629	T111	J55	6	625	187	22	submit	-
629	T111	J55	6	625	187	22	start	M02
645	T048	J24	0	3125	466	310	finish	M01
645	T092	J53	0	937	277	53	start	M01
645	T093	J53	0	937	277	54	start	M01
647	T049	J25	6	937	1244	298	finish	M01
647	T094	J53	0	937	277	53	start	M01
647	T095	J53	0	937	277	64	start	M01
651	T111	J55	6	625	187	22	finish	M02
655	T046	J22	6	625	280	358	finish	M01

time	task	job	priority	CPU	memory	run time	event	machine
655	T096	J53	0	937	277	63	start	M01
662	T091	J53	0	937	277	54	finish	M05
662	T100	J53	0	937	277	51	start	M05
675	T112	J56	6	625	280	118	submit	-
675	T112	J56	6	625	280	118	start	M02
678	T090	J52	0	1250	777	71	finish	M08
678	T097	J53	0	937	277	67	start	M08
678	T098	J53	0	937	277	53	start	M08
680	T087	J51	1	937	777	75	finish	M05
680	T088	J52	0	1250	777	73	finish	M06
680	T089	J52	0	1250	777	73	finish	M07
680	T099	J53	0	937	277	63	start	M06
680	T101	J53	0	937	277	54	start	M06
680	T102	J53	0	937	277	57	start	M07
680	T103	J53	0	937	277	55	start	M07
689	T113	J57	6	625	187	15	submit	-
689	T113	J57	6	625	187	15	start	M05
695	T114	J58	9	6250	233	115	submit	-
695	T096	J53	0	937	277	63	evict	M01
695	T095	J53	0	937	277	64	evict	M01
695	T094	J53	0	937	277	53	evict	M01
695	T093	J53	0	937	277	54	evict	M01
695	T092	J53	0	937	277	53	evict	M01
695	T047	J23	1	1250	777	1057	evict	M01
695	T114	J58	9	6250	233	115	start	M01
697	T115	J59	0	6250	466	34	submit	-
704	T113	J57	6	625	187	15	finish	M05
706	T116	J60	6	937	1244	71	submit	-
706	T117	J61	4	625	155	23	submit	-
706	T118	J62	8	625	466	1395	submit	-
706	T119	J63	4	1250	398	19	submit	-
706	T120	J64	4	625	311	13	submit	-
710	T121	J65	9	6250	1590	80	submit	-
713	T069	J39	6	4376	777	241	finish	M05
713	T100	J53	0	937	277	51	finish	M05
713	T121	J65	9	6250	1590	80	start	M05
722	T122	J66	4	3125	1125	66	submit	-
731	T098	J53	0	937	277	53	finish	M08
731	T118	J62	8	625	466	1395	start	M08
733	T071	J39	6	4376	777	261	finish	M07
734	T101	J53	0	937	277	54	finish	M06
734	T116	J60	6	937	1244	71	start	M06
735	T103	J53	0	937	277	55	finish	M07
737	T102	J53	0	937	277	57	finish	M07
737	T086	J50	8	6250	1590	3241	start	M07
742	T070	J39	6	4376	777	270	finish	M06
742	T074	J40	4	3448	1460	561	start	M06
742	T117	J61	4	625	155	23	start	M06
743	T099	J53	0	937	277	63	finish	M06
743	T120	J64	4	625	311	13	start	M06
745	T097	J53	0	937	277	67	finish	M08
745	T092	J53	0	937	277	53	start	M08
756	T081	J45	6	937	1244	184	finish	M03
756	T120	J64	4	625	311	13	finish	M06
756	T093	J53	0	937	277	54	start	M03
756	T094	J53	0	937	277	53	start	M06
762	T123	J67	6	625	280	223	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
763	T124	J68	4	937	549	806	submit	-
763	T125	J68	4	937	549	809	submit	-
763	T126	J68	4	937	549	797	submit	-
763	T127	J68	4	937	549	807	submit	-
765	T117	J61	4	625	155	23	finish	M06
765	T123	J67	6	625	280	223	start	M06
771	T079	J43	6	625	280	229	finish	M02
789	T128	J69	6	937	1244	185	submit	-
789	T129	J70	6	625	280	107	submit	-
791	T130	J71	6	1250	637	184	submit	-
791	T131	J71	6	1250	637	164	submit	-
791	T132	J71	6	1250	637	181	submit	-
791	T133	J71	6	1250	637	166	submit	-
791	T134	J72	6	937	255	55	submit	-
791	T135	J72	6	937	255	57	submit	-
791	T136	J72	6	937	255	135	submit	-
791	T137	J72	6	937	255	123	submit	-
793	T112	J56	6	625	280	118	finish	M02
793	T121	J65	9	6250	1590	80	finish	M05
793	T128	J69	6	937	1244	185	start	M02
793	T129	J70	6	625	280	107	start	M05
793	T130	J71	6	1250	637	184	start	M05
793	T131	J71	6	1250	637	164	start	M05
793	T132	J71	6	1250	637	181	start	M05
793	T133	J71	6	1250	637	166	start	M05
798	T092	J53	0	937	277	53	finish	M08
798	T134	J72	6	937	255	55	start	M08
804	T138	J73	6	625	3180	27	submit	-
804	T138	J73	6	625	3180	27	start	M05
805	T116	J60	6	937	1244	71	finish	M06
805	T135	J72	6	937	255	57	start	M06
809	T094	J53	0	937	277	53	finish	M06
809	T136	J72	6	937	255	135	start	M06
810	T093	J53	0	937	277	54	finish	M03
810	T114	J58	9	6250	233	115	finish	M01
810	T137	J72	6	937	255	123	start	M01
810	T075	J40	4	3448	1460	521	start	M01
810	T119	J63	4	1250	398	19	start	M01
810	T124	J68	4	937	549	806	start	M03
819	T139	J74	6	937	1244	68	submit	-
824	T072	J39	6	4376	777	352	finish	M08
824	T139	J74	6	937	1244	68	start	M04
824	T076	J40	4	3448	1460	595	start	M04
824	T077	J41	4	3125	1224	42	start	M08
824	T125	J68	4	937	549	809	start	M04
824	T126	J68	4	937	549	797	start	M08
829	T119	J63	4	1250	398	19	finish	M01
829	T127	J68	4	937	549	807	start	M01
831	T138	J73	6	625	3180	27	finish	M05
853	T134	J72	6	937	255	55	finish	M08
853	T047	J23	1	1250	777	1057	start	M08
862	T135	J72	6	937	255	57	finish	M06
862	T095	J53	0	937	277	64	start	M06
866	T077	J41	4	3125	1224	42	finish	M08
866	T085	J49	4	3125	1125	26	start	M08
882	T140	J75	6	6250	622	115	submit	-
882	T141	J76	6	625	280	55	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
882	T141	J76	6	625	280	55	start	M01
888	T142	J77	1	937	622	236	submit	-
890	T143	J78	0	1250	622	130	submit	-
890	T144	J78	0	1250	622	116	submit	-
890	T145	J78	0	1250	622	115	submit	-
890	T146	J78	0	1250	622	141	submit	-
890	T147	J78	0	1250	622	195	submit	-
890	T148	J78	0	1250	622	139	submit	-
890	T149	J78	0	1250	622	162	submit	-
890	T150	J78	0	1250	622	176	submit	-
890	T151	J78	0	1250	622	117	submit	-
890	T152	J78	0	1250	622	119	submit	-
890	T153	J78	0	1250	622	157	submit	-
890	T154	J78	0	1250	622	119	submit	-
890	T155	J78	0	1250	622	170	submit	-
890	T156	J78	0	1250	622	119	submit	-
890	T157	J78	0	1250	622	180	submit	-
890	T158	J78	0	1250	622	126	submit	-
892	T085	J49	4	3125	1125	26	finish	M08
892	T139	J74	6	937	1244	68	finish	M04
892	T122	J66	4	3125	1125	66	start	M08
892	T142	J77	1	937	622	236	start	M04
900	T129	J70	6	625	280	107	finish	M05
900	T096	J53	0	937	277	63	start	M05
914	T159	J79	6	625	3180	31	submit	-
914	T159	J79	6	625	3180	31	start	M04
926	T095	J53	0	937	277	64	finish	M06
926	T104	J53	0	937	277	55	start	M06
933	T137	J72	6	937	255	123	finish	M01
933	T105	J53	0	937	277	51	start	M01
937	T141	J76	6	625	280	55	finish	M01
944	T136	J72	6	937	255	135	finish	M06
944	T106	J53	0	937	277	51	start	M06
945	T159	J79	6	625	3180	31	finish	M04
948	T160	J80	4	62	398	17	submit	-
948	T160	J80	4	62	398	17	start	M01
957	T131	J71	6	1250	637	164	finish	M05
958	T122	J66	4	3125	1125	66	finish	M08
958	T143	J78	0	1250	622	130	start	M08
958	T144	J78	0	1250	622	116	start	M08
959	T133	J71	6	1250	637	166	finish	M05
963	T096	J53	0	937	277	63	finish	M05
965	T160	J80	4	62	398	17	finish	M01
974	T132	J71	6	1250	637	181	finish	M05
977	T130	J71	6	1250	637	184	finish	M05
977	T140	J75	6	6250	622	115	start	M05
978	T128	J69	6	937	1244	185	finish	M02
978	T145	J78	0	1250	622	115	start	M02
981	T104	J53	0	937	277	55	finish	M06
984	T105	J53	0	937	277	51	finish	M01
984	T146	J78	0	1250	622	141	start	M01
988	T123	J67	6	625	280	223	finish	M06
988	T147	J78	0	1250	622	195	start	M06
995	T106	J53	0	937	277	51	finish	M06
995	T148	J78	0	1250	622	139	start	M06
997	T161	J81	6	625	280	129	submit	-
997	T161	J81	6	625	280	129	start	M04

time	task	job	priority	CPU	memory	run time	event	machine
1035	T162	J82	6	625	280	262	submit	-
1035	T163	J83	6	937	1244	78	submit	-
1035	T164	J84	4	625	155	36	submit	-
1036	T165	J85	6	937	197	74	submit	-
1036	T166	J85	6	937	197	73	submit	-
1036	T167	J85	6	937	197	74	submit	-
1036	T168	J85	6	937	197	74	submit	-
1057	T169	J86	0	6250	1244	53	submit	-
1060	T170	J87	0	6250	1244	51	submit	-
1074	T144	J78	0	1250	622	116	finish	M08
1074	T162	J82	6	625	280	262	start	M08
1074	T165	J85	6	937	197	74	start	M08
1074	T164	J84	4	625	155	36	start	M08
1083	T171	J88	6	937	1244	200	submit	-
1084	T172	J89	6	1250	633	153	submit	-
1084	T173	J89	6	1250	633	165	submit	-
1084	T174	J89	6	1250	633	153	submit	-
1084	T175	J89	6	1250	633	171	submit	-
1084	T176	J90	6	937	258	199	submit	-
1084	T177	J90	6	937	258	113	submit	-
1084	T178	J90	6	937	258	46	submit	-
1084	T179	J90	6	937	258	42	submit	-
1088	T143	J78	0	1250	622	130	finish	M08
1088	T166	J85	6	937	197	73	start	M08
1092	T140	J75	6	6250	622	115	finish	M05
1092	T163	J83	6	937	1244	78	start	M05
1092	T167	J85	6	937	197	74	start	M05
1092	T168	J85	6	937	197	74	start	M05
1092	T171	J88	6	937	1244	200	start	M05
1092	T172	J89	6	1250	633	153	start	M05
1092	T173	J89	6	1250	633	165	start	M05
1093	T145	J78	0	1250	622	115	finish	M02
1093	T174	J89	6	1250	633	153	start	M02
1109	T180	J91	1	937	777	86	submit	-
1110	T164	J84	4	625	155	36	finish	M08
1110	T176	J90	6	937	258	199	start	M08
1111	T181	J92	0	1250	777	83	submit	-
1111	T182	J92	0	1250	777	80	submit	-
1111	T183	J92	0	1250	777	82	submit	-
1111	T184	J92	0	1250	777	84	submit	-
1111	T185	J92	0	1250	777	83	submit	-
1111	T186	J92	0	1250	777	74	submit	-
1111	T187	J92	0	1250	777	73	submit	-
1111	T188	J92	0	1250	777	82	submit	-
1111	T189	J92	0	1250	777	81	submit	-
1111	T190	J92	0	1250	777	71	submit	-
1113	T191	J93	0	937	277	58	submit	-
1113	T192	J93	0	937	277	49	submit	-
1113	T193	J93	0	937	277	50	submit	-
1113	T194	J93	0	937	277	48	submit	-
1113	T195	J93	0	937	277	47	submit	-
1113	T196	J93	0	937	277	49	submit	-
1113	T197	J93	0	937	277	74	submit	-
1113	T198	J93	0	937	277	58	submit	-
1113	T199	J93	0	937	277	52	submit	-
1113	T200	J93	0	937	277	47	submit	-
1113	T201	J93	0	937	277	51	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
1113	T202	J93	0	937	277	63	submit	-
1113	T203	J93	0	937	277	47	submit	-
1113	T204	J93	0	937	277	46	submit	-
1113	T205	J93	0	937	277	48	submit	-
1113	T206	J93	0	937	277	47	submit	-
1114	T073	J40	4	3448	1460	599	finish	M02
1114	T175	J89	6	1250	633	171	start	M02
1114	T177	J90	6	937	258	113	start	M02
1114	T178	J90	6	937	258	46	start	M02
1125	T146	J78	0	1250	622	141	finish	M01
1125	T179	J90	6	937	258	42	start	M01
1126	T161	J81	6	625	280	129	finish	M04
1128	T142	J77	1	937	622	236	finish	M04
1128	T180	J91	1	937	777	86	start	M04
1134	T148	J78	0	1250	622	139	finish	M06
1134	T149	J78	0	1250	622	162	start	M06
1135	T207	J94	4	6250	2188	16	submit	-
1148	T165	J85	6	937	197	74	finish	M08
1148	T191	J93	0	937	277	58	start	M08
1160	T178	J90	6	937	258	46	finish	M02
1160	T150	J78	0	1250	622	176	start	M02
1161	T166	J85	6	937	197	73	finish	M08
1161	T192	J93	0	937	277	49	start	M08
1166	T167	J85	6	937	197	74	finish	M05
1166	T168	J85	6	937	197	74	finish	M05
1166	T186	J92	0	1250	777	74	start	M05
1167	T179	J90	6	937	258	42	finish	M01
1167	T151	J78	0	1250	622	117	start	M01
1170	T163	J83	6	937	1244	78	finish	M05
1170	T187	J92	0	1250	777	73	start	M05
1183	T147	J78	0	1250	622	195	finish	M06
1183	T152	J78	0	1250	622	119	start	M06

Cell window 23: Events

Time 416: machine M04 failed; no task was running on it.

Time 695: task T114 (priority 9) evicted task T047 (priority 1) from machine M01.

Time 695: task T114 (priority 9) evicted task T092 (priority 0) from machine M01.

Time 695: task T114 (priority 9) evicted task T093 (priority 0) from machine M01.

Time 695: task T114 (priority 9) evicted task T094 (priority 0) from machine M01.

Time 695: task T114 (priority 9) evicted task T095 (priority 0) from machine M01.

Time 695: task T114 (priority 9) evicted task T096 (priority 0) from machine M01.

Time 745: task T092 restarted on machine M08.

Time 756: task T093 restarted on machine M03.

Time 756: task T094 restarted on machine M06.

Time 824: machine M04 returned.

Time 853: task T047 restarted on machine M08.

Time 862: task T095 restarted on machine M06.

Time 900: task T096 restarted on machine M05.

Longest waits (submission to first start, tasks started by the window end): task T106 waited 336 s (submitted at 608, first started at 944); task T105 waited 325 s (submitted at 608, first started at 933); task T104 waited 318 s (submitted at 608, first started at 926); task T076 waited 309 s (submitted at 515, first started at 824); task T075 waited 295 s (submitted at 515, first started at 810).

Cell window 23: Queue at window end

State after the events at time 1200.

Tasks running at time 1200 (remaining run time = seconds still to run after time 1200):

task	job	priority	CPU	memory	machine	started	remaining
T075	J40	4	3448	1460	M01	810	131
T127	J68	4	937	549	M01	829	436
T151	J78	0	1250	622	M01	1167	84
T078	J42	1	1250	777	M02	538	432
T150	J78	0	1250	622	M02	1160	136
T174	J89	6	1250	633	M02	1093	46
T175	J89	6	1250	633	M02	1114	85
T177	J90	6	937	258	M02	1114	27
T055	J28	0	1688	295	M03	366	162
T068	J38	8	1874	777	M03	456	2669
T084	J48	0	1688	295	M03	586	420
T124	J68	4	937	549	M03	810	416
T076	J40	4	3448	1460	M04	824	219
T125	J68	4	937	549	M04	824	433
T180	J91	1	937	777	M04	1128	14
T171	J88	6	937	1244	M05	1092	92
T172	J89	6	1250	633	M05	1092	45
T173	J89	6	1250	633	M05	1092	57
T186	J92	0	1250	777	M05	1166	40
T187	J92	0	1250	777	M05	1170	43
T074	J40	4	3448	1460	M06	742	103
T149	J78	0	1250	622	M06	1134	96
T152	J78	0	1250	622	M06	1183	102
T086	J50	8	6250	1590	M07	737	2778
T047	J23	1	1250	777	M08	853	710
T118	J62	8	625	466	M08	731	926
T126	J68	4	937	549	M08	824	421
T162	J82	6	625	280	M08	1074	136
T176	J90	6	937	258	M08	1110	109
T191	J93	0	937	277	M08	1148	6
T192	J93	0	937	277	M08	1161	10

Tasks waiting at time 1200, in queue order (each will run its full run time once started):

task	job	priority	CPU	memory	submitted	run time
T107	J54	4	3125	3180	614	1894
T108	J54	4	3125	3180	614	1800
T109	J54	4	3125	3180	614	1761
T110	J54	4	3125	3180	614	2562
T207	J94	4	6250	2188	1135	16
T082	J46	0	6250	1244	575	62
T083	J47	0	6250	1244	576	83
T115	J59	0	6250	466	697	34
T153	J78	0	1250	622	890	157
T154	J78	0	1250	622	890	119
T155	J78	0	1250	622	890	170
T156	J78	0	1250	622	890	119

task	job	priority	CPU	memory	submitted	run time
T157	J78	0	1250	622	890	180
T158	J78	0	1250	622	890	126
T169	J86	0	6250	1244	1057	53
T170	J87	0	6250	1244	1060	51
T181	J92	0	1250	777	1111	83
T182	J92	0	1250	777	1111	80
T183	J92	0	1250	777	1111	82
T184	J92	0	1250	777	1111	84
T185	J92	0	1250	777	1111	83
T188	J92	0	1250	777	1111	82
T189	J92	0	1250	777	1111	81
T190	J92	0	1250	777	1111	71
T193	J93	0	937	277	1113	50
T194	J93	0	937	277	1113	48
T195	J93	0	937	277	1113	47
T196	J93	0	937	277	1113	49
T197	J93	0	937	277	1113	74
T198	J93	0	937	277	1113	58
T199	J93	0	937	277	1113	52
T200	J93	0	937	277	1113	47
T201	J93	0	937	277	1113	51
T202	J93	0	937	277	1113	63
T203	J93	0	937	277	1113	47
T204	J93	0	937	277	1113	46
T205	J93	0	937	277	1113	48
T206	J93	0	937	277	1113	47

All machines are up at time 1200.

Cell window 26

Times 0 to 1200. The cluster has 8 machines. 137 tasks of 52 jobs are submitted during the window.

Cell window 26: Machines

machine	CPU	memory	fails at	returns at
M01	6250	3116	-	-
M02	6250	6244	544	1207
M03	6250	3116	-	-
M04	6250	6244	-	-
M05	6250	9363	-	-
M06	12500	12500	-	-
M07	6250	3116	-	-
M08	6250	6244	-	-

Cell window 26: Task log

time	task	job	priority	CPU	memory	run time	event	machine
11	T001	J01	6	625	280	107	submit	-
11	T001	J01	6	625	280	107	start	M01
13	T002	J02	6	937	255	55	submit	-
13	T003	J02	6	937	255	57	submit	-
13	T004	J02	6	937	255	135	submit	-
13	T005	J02	6	937	255	123	submit	-
13	T002	J02	6	937	255	55	start	M01
13	T003	J02	6	937	255	57	start	M01
13	T004	J02	6	937	255	135	start	M01
13	T005	J02	6	937	255	123	start	M01
37	T006	J03	9	4688	6360	491	submit	-
37	T006	J03	9	4688	6360	491	start	M05
68	T002	J02	6	937	255	55	finish	M01
70	T003	J02	6	937	255	57	finish	M01
104	T007	J04	6	625	187	36	submit	-
104	T007	J04	6	625	187	36	start	M01
110	T008	J05	1	937	622	236	submit	-
110	T008	J05	1	937	622	236	start	M01
112	T009	J06	0	1250	622	130	submit	-
112	T010	J06	0	1250	622	116	submit	-
112	T011	J06	0	1250	622	115	submit	-
112	T012	J06	0	1250	622	141	submit	-
112	T013	J06	0	1250	622	195	submit	-
112	T014	J06	0	1250	622	139	submit	-
112	T015	J06	0	1250	622	162	submit	-
112	T016	J06	0	1250	622	176	submit	-
112	T017	J06	0	1250	622	117	submit	-
112	T018	J06	0	1250	622	119	submit	-
112	T019	J06	0	1250	622	157	submit	-
112	T020	J06	0	1250	622	119	submit	-
112	T021	J06	0	1250	622	170	submit	-
112	T022	J06	0	1250	622	119	submit	-
112	T023	J06	0	1250	622	180	submit	-
112	T024	J06	0	1250	622	126	submit	-
112	T009	J06	0	1250	622	130	start	M01
112	T010	J06	0	1250	622	116	start	M02
112	T011	J06	0	1250	622	115	start	M02
112	T012	J06	0	1250	622	141	start	M02

time	task	job	priority	CPU	memory	run time	event	machine
112	T013	J06	0	1250	622	195	start	M02
112	T014	J06	0	1250	622	139	start	M02
112	T015	J06	0	1250	622	162	start	M03
112	T016	J06	0	1250	622	176	start	M03
112	T017	J06	0	1250	622	117	start	M03
112	T018	J06	0	1250	622	119	start	M03
112	T019	J06	0	1250	622	157	start	M03
112	T020	J06	0	1250	622	119	start	M04
112	T021	J06	0	1250	622	170	start	M04
112	T022	J06	0	1250	622	119	start	M04
112	T023	J06	0	1250	622	180	start	M04
112	T024	J06	0	1250	622	126	start	M04
118	T001	J01	6	625	280	107	finish	M01
136	T005	J02	6	937	255	123	finish	M01
140	T007	J04	6	625	187	36	finish	M01
148	T004	J02	6	937	255	135	finish	M01
170	T025	J07	4	62	398	17	submit	-
170	T025	J07	4	62	398	17	start	M01
187	T025	J07	4	62	398	17	finish	M01
227	T011	J06	0	1250	622	115	finish	M02
228	T010	J06	0	1250	622	116	finish	M02
229	T017	J06	0	1250	622	117	finish	M03
231	T018	J06	0	1250	622	119	finish	M03
231	T020	J06	0	1250	622	119	finish	M04
231	T022	J06	0	1250	622	119	finish	M04
238	T024	J06	0	1250	622	126	finish	M04
242	T009	J06	0	1250	622	130	finish	M01
251	T014	J06	0	1250	622	139	finish	M02
253	T012	J06	0	1250	622	141	finish	M02
257	T026	J08	4	625	155	36	submit	-
257	T026	J08	4	625	155	36	start	M01
258	T027	J09	6	645	197	74	submit	-
258	T028	J09	6	645	197	76	submit	-
258	T027	J09	6	645	197	74	start	M01
258	T028	J09	6	645	197	76	start	M01
269	T019	J06	0	1250	622	157	finish	M03
274	T015	J06	0	1250	622	162	finish	M03
279	T029	J10	0	6250	1244	53	submit	-
279	T029	J10	0	6250	1244	53	start	M06
282	T021	J06	0	1250	622	170	finish	M04
282	T030	J11	0	6250	1244	51	submit	-
282	T030	J11	0	6250	1244	51	start	M06
288	T016	J06	0	1250	622	176	finish	M03
292	T023	J06	0	1250	622	180	finish	M04
293	T026	J08	4	625	155	36	finish	M01
307	T013	J06	0	1250	622	195	finish	M02
331	T031	J12	1	937	777	86	submit	-
331	T031	J12	1	937	777	86	start	M01
332	T027	J09	6	645	197	74	finish	M01
332	T029	J10	0	6250	1244	53	finish	M06
333	T030	J11	0	6250	1244	51	finish	M06
333	T032	J13	0	1250	777	83	submit	-
333	T033	J13	0	1250	777	80	submit	-
333	T034	J13	0	1250	777	82	submit	-
333	T035	J13	0	1250	777	84	submit	-
333	T036	J13	0	1250	777	83	submit	-
333	T037	J13	0	1250	777	74	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
333	T038	J13	0	1250	777	73	submit	-
333	T039	J13	0	1250	777	82	submit	-
333	T040	J13	0	1250	777	81	submit	-
333	T041	J13	0	1250	777	71	submit	-
333	T032	J13	0	1250	777	83	start	M01
333	T033	J13	0	1250	777	80	start	M02
333	T034	J13	0	1250	777	82	start	M02
333	T035	J13	0	1250	777	84	start	M02
333	T036	J13	0	1250	777	83	start	M02
333	T037	J13	0	1250	777	74	start	M02
333	T038	J13	0	1250	777	73	start	M03
333	T039	J13	0	1250	777	82	start	M03
333	T040	J13	0	1250	777	81	start	M03
333	T041	J13	0	1250	777	71	start	M03
334	T028	J09	6	645	197	76	finish	M01
335	T042	J14	0	937	277	58	submit	-
335	T043	J14	0	937	277	49	submit	-
335	T044	J14	0	937	277	50	submit	-
335	T045	J14	0	937	277	48	submit	-
335	T046	J14	0	937	277	47	submit	-
335	T047	J14	0	937	277	49	submit	-
335	T048	J14	0	937	277	74	submit	-
335	T049	J14	0	937	277	58	submit	-
335	T050	J14	0	937	277	52	submit	-
335	T051	J14	0	937	277	47	submit	-
335	T052	J14	0	937	277	51	submit	-
335	T053	J14	0	937	277	63	submit	-
335	T054	J14	0	937	277	47	submit	-
335	T055	J14	0	937	277	46	submit	-
335	T056	J14	0	937	277	48	submit	-
335	T057	J14	0	937	277	47	submit	-
335	T042	J14	0	937	277	58	start	M01
335	T043	J14	0	937	277	49	start	M01
335	T044	J14	0	937	277	50	start	M01
335	T045	J14	0	937	277	48	start	M04
335	T046	J14	0	937	277	47	start	M04
335	T047	J14	0	937	277	49	start	M04
335	T048	J14	0	937	277	74	start	M04
335	T049	J14	0	937	277	58	start	M04
335	T050	J14	0	937	277	52	start	M04
335	T051	J14	0	937	277	47	start	M05
335	T052	J14	0	937	277	51	start	M06
335	T053	J14	0	937	277	63	start	M06
335	T054	J14	0	937	277	47	start	M06
335	T055	J14	0	937	277	46	start	M06
335	T056	J14	0	937	277	48	start	M06
335	T057	J14	0	937	277	47	start	M06
346	T008	J05	1	937	622	236	finish	M01
381	T055	J14	0	937	277	46	finish	M06
382	T046	J14	0	937	277	47	finish	M04
382	T051	J14	0	937	277	47	finish	M05
382	T054	J14	0	937	277	47	finish	M06
382	T057	J14	0	937	277	47	finish	M06
383	T045	J14	0	937	277	48	finish	M04
383	T056	J14	0	937	277	48	finish	M06
384	T043	J14	0	937	277	49	finish	M01
384	T047	J14	0	937	277	49	finish	M04

time	task	job	priority	CPU	memory	run time	event	machine
385	T044	J14	0	937	277	50	finish	M01
386	T052	J14	0	937	277	51	finish	M06
387	T050	J14	0	937	277	52	finish	M04
393	T042	J14	0	937	277	58	finish	M01
393	T049	J14	0	937	277	58	finish	M04
398	T053	J14	0	937	277	63	finish	M06
404	T041	J13	0	1250	777	71	finish	M03
406	T038	J13	0	1250	777	73	finish	M03
407	T037	J13	0	1250	777	74	finish	M02
409	T048	J14	0	937	277	74	finish	M04
413	T033	J13	0	1250	777	80	finish	M02
414	T040	J13	0	1250	777	81	finish	M03
415	T034	J13	0	1250	777	82	finish	M02
415	T039	J13	0	1250	777	82	finish	M03
416	T032	J13	0	1250	777	83	finish	M01
416	T036	J13	0	1250	777	83	finish	M02
417	T031	J12	1	937	777	86	finish	M01
417	T035	J13	0	1250	777	84	finish	M02
458	T058	J15	0	3125	466	266	submit	-
458	T058	J15	0	3125	466	266	start	M01
518	T059	J16	0	6250	466	38	submit	-
518	T059	J16	0	6250	466	38	start	M02
519	T060	J17	1	1250	1590	374	submit	-
519	T061	J18	4	1250	466	275	submit	-
519	T062	J18	4	1250	466	257	submit	-
519	T063	J18	4	1250	466	257	submit	-
519	T064	J18	4	1250	466	243	submit	-
519	T061	J18	4	1250	466	275	start	M01
519	T062	J18	4	1250	466	257	start	M01
519	T063	J18	4	1250	466	257	start	M03
519	T064	J18	4	1250	466	243	start	M03
519	T060	J17	1	1250	1590	374	start	M03
520	T065	J19	1	2499	311	252	submit	-
520	T065	J19	1	2499	311	252	start	M03
528	T006	J03	9	4688	6360	491	finish	M05
530	T066	J20	9	6250	1590	79	submit	-
530	T066	J20	9	6250	1590	79	start	M04
534	T067	J21	4	62	398	18	submit	-
534	T068	J22	4	625	311	25	submit	-
534	T069	J23	4	625	155	29	submit	-
534	T067	J21	4	62	398	18	start	M01
534	T068	J22	4	625	311	25	start	M05
534	T069	J23	4	625	155	29	start	M05
544	T059	J16	0	6250	466	38	fail	M02
544	T059	J16	0	6250	466	38	start	M06
552	T067	J21	4	62	398	18	finish	M01
553	T070	J24	1	1250	466	175	submit	-
553	T070	J24	1	1250	466	175	start	M05
559	T068	J22	4	625	311	25	finish	M05
562	T071	J25	1	3748	466	197	submit	-
562	T071	J25	1	3748	466	197	start	M05
563	T069	J23	4	625	155	29	finish	M05
565	T072	J26	6	645	197	72	submit	-
565	T073	J27	0	1524	271	174	submit	-
565	T074	J27	0	1524	271	175	submit	-
565	T075	J27	0	1524	271	175	submit	-
565	T076	J27	0	1524	271	177	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
565	T077	J27	0	1524	271	175	submit	-
565	T078	J27	0	1524	271	177	submit	-
565	T079	J27	0	1524	271	173	submit	-
565	T080	J27	0	1524	271	175	submit	-
565	T081	J27	0	1524	271	173	submit	-
565	T082	J27	0	1524	271	176	submit	-
565	T083	J27	0	1524	271	175	submit	-
565	T084	J27	0	1524	271	174	submit	-
565	T085	J27	0	1524	271	176	submit	-
565	T086	J27	0	1524	271	173	submit	-
565	T087	J27	0	1524	271	174	submit	-
565	T088	J27	0	1524	271	176	submit	-
565	T072	J26	6	645	197	72	start	M05
565	T073	J27	0	1524	271	174	start	M06
565	T074	J27	0	1524	271	175	start	M06
565	T075	J27	0	1524	271	175	start	M06
565	T076	J27	0	1524	271	177	start	M06
565	T077	J27	0	1524	271	175	start	M07
565	T078	J27	0	1524	271	177	start	M07
565	T079	J27	0	1524	271	173	start	M07
565	T080	J27	0	1524	271	175	start	M07
565	T081	J27	0	1524	271	173	start	M08
565	T082	J27	0	1524	271	176	start	M08
565	T083	J27	0	1524	271	175	start	M08
565	T084	J27	0	1524	271	174	start	M08
582	T059	J16	0	6250	466	38	finish	M06
582	T085	J27	0	1524	271	176	start	M06
582	T086	J27	0	1524	271	173	start	M06
582	T087	J27	0	1524	271	174	start	M06
582	T088	J27	0	1524	271	176	start	M06
606	T089	J28	1	1250	777	1139	submit	-
609	T066	J20	9	6250	1590	79	finish	M04
609	T089	J28	1	1250	777	1139	start	M04
637	T072	J26	6	645	197	72	finish	M05
639	T090	J29	9	4688	6360	487	submit	-
639	T071	J25	1	3748	466	197	evict	M05
639	T090	J29	9	4688	6360	487	start	M05
639	T071	J25	1	3748	466	197	start	M04
640	T091	J30	1	625	622	783	submit	-
640	T091	J30	1	625	622	783	start	M01
649	T092	J31	4	6873	1910	2931	submit	-
649	T093	J31	4	6873	1910	2911	submit	-
649	T094	J31	4	6873	1910	2994	submit	-
649	T095	J31	4	6873	1910	2907	submit	-
658	T096	J32	0	1688	295	1070	submit	-
724	T058	J15	0	3125	466	266	finish	M01
724	T096	J32	0	1688	295	1070	start	M01
728	T070	J24	1	1250	466	175	finish	M05
738	T079	J27	0	1524	271	173	finish	M07
738	T081	J27	0	1524	271	173	finish	M08
739	T073	J27	0	1524	271	174	finish	M06
739	T084	J27	0	1524	271	174	finish	M08
740	T074	J27	0	1524	271	175	finish	M06
740	T075	J27	0	1524	271	175	finish	M06
740	T077	J27	0	1524	271	175	finish	M07
740	T080	J27	0	1524	271	175	finish	M07
740	T083	J27	0	1524	271	175	finish	M08

time	task	job	priority	CPU	memory	run time	event	machine
741	T082	J27	0	1524	271	176	finish	M08
742	T076	J27	0	1524	271	177	finish	M06
742	T078	J27	0	1524	271	177	finish	M07
755	T086	J27	0	1524	271	173	finish	M06
755	T092	J31	4	6873	1910	2931	start	M06
756	T087	J27	0	1524	271	174	finish	M06
757	T097	J33	0	6250	1244	33	submit	-
757	T097	J33	0	6250	1244	33	start	M07
758	T085	J27	0	1524	271	176	finish	M06
758	T088	J27	0	1524	271	176	finish	M06
758	T098	J34	0	6250	1244	68	submit	-
758	T098	J34	0	6250	1244	68	start	M08
762	T064	J18	4	1250	466	243	finish	M03
772	T065	J19	1	2499	311	252	finish	M03
773	T099	J35	1	625	16	739	submit	-
773	T099	J35	1	625	16	739	start	M01
776	T062	J18	4	1250	466	257	finish	M01
776	T063	J18	4	1250	466	257	finish	M03
780	T100	J36	1	4376	777	634	submit	-
780	T100	J36	1	4376	777	634	start	M03
790	T097	J33	0	6250	1244	33	finish	M07
794	T061	J18	4	1250	466	275	finish	M01
819	T101	J37	0	1250	16	10	submit	-
819	T101	J37	0	1250	16	10	start	M01
822	T102	J38	9	6250	233	110	submit	-
822	T102	J38	9	6250	233	110	start	M07
825	T103	J39	1	3125	777	576	submit	-
825	T103	J39	1	3125	777	576	start	M06
826	T098	J34	0	6250	1244	68	finish	M08
827	T104	J40	8	3125	388	20	submit	-
827	T105	J41	4	6873	1910	2936	submit	-
827	T106	J41	4	6873	1910	2954	submit	-
827	T107	J41	4	6873	1910	3065	submit	-
827	T108	J41	4	6873	1910	2937	submit	-
827	T109	J42	0	1494	311	573	submit	-
827	T104	J40	8	3125	388	20	start	M08
827	T109	J42	0	1494	311	573	start	M01
828	T110	J43	0	625	311	360	submit	-
828	T111	J43	0	625	311	346	submit	-
828	T112	J43	0	625	311	346	submit	-
828	T113	J43	0	625	311	368	submit	-
828	T114	J43	0	625	311	409	submit	-
828	T115	J43	0	625	311	336	submit	-
828	T116	J43	0	625	311	348	submit	-
828	T117	J43	0	625	311	573	submit	-
828	T118	J43	0	625	311	348	submit	-
828	T119	J43	0	625	311	337	submit	-
828	T120	J43	0	625	311	418	submit	-
828	T121	J43	0	625	311	375	submit	-
828	T122	J43	0	625	311	334	submit	-
828	T123	J43	0	625	311	359	submit	-
828	T124	J43	0	625	311	437	submit	-
828	T125	J43	0	625	311	378	submit	-
828	T110	J43	0	625	311	360	start	M04
828	T111	J43	0	625	311	346	start	M04
828	T112	J43	0	625	311	346	start	M05
828	T113	J43	0	625	311	368	start	M05

time	task	job	priority	CPU	memory	run time	event	machine
828	T114	J43	0	625	311	409	start	M06
828	T115	J43	0	625	311	336	start	M06
828	T116	J43	0	625	311	348	start	M06
828	T117	J43	0	625	311	573	start	M06
828	T118	J43	0	625	311	348	start	M08
828	T119	J43	0	625	311	337	start	M08
828	T120	J43	0	625	311	418	start	M08
828	T121	J43	0	625	311	375	start	M08
828	T122	J43	0	625	311	334	start	M08
829	T101	J37	0	1250	16	10	finish	M01
829	T123	J43	0	625	311	359	start	M01
829	T124	J43	0	625	311	437	start	M01
836	T071	J25	1	3748	466	197	finish	M04
836	T125	J43	0	625	311	378	start	M04
847	T104	J40	8	3125	388	20	finish	M08
865	T126	J44	1	1250	777	1121	submit	-
865	T126	J44	1	1250	777	1121	start	M04
893	T060	J17	1	1250	1590	374	finish	M03
894	T127	J45	8	1250	2863	2887	submit	-
894	T127	J45	8	1250	2863	2887	start	M04
908	T128	J46	0	1688	295	1065	submit	-
908	T128	J46	0	1688	295	1065	start	M03
932	T102	J38	9	6250	233	110	finish	M07
1067	T129	J47	4	62	398	18	submit	-
1067	T129	J47	4	62	398	18	start	M01
1085	T129	J47	4	62	398	18	finish	M01
1093	T130	J48	8	625	155	213	submit	-
1093	T130	J48	8	625	155	213	start	M04
1108	T131	J49	4	6873	1910	3002	submit	-
1108	T132	J49	4	6873	1910	2924	submit	-
1108	T133	J49	4	6873	1910	2901	submit	-
1108	T134	J49	4	6873	1910	2924	submit	-
1119	T135	J50	0	6250	466	39	submit	-
1119	T135	J50	0	6250	466	39	start	M07
1122	T136	J51	4	625	155	27	submit	-
1122	T136	J51	4	625	155	27	start	M08
1126	T090	J29	9	4688	6360	487	finish	M05
1128	T137	J52	9	6250	1590	70	submit	-
1128	T124	J43	0	625	311	437	evict	M01
1128	T123	J43	0	625	311	359	evict	M01
1128	T109	J42	0	1494	311	573	evict	M01
1128	T096	J32	0	1688	295	1070	evict	M01
1128	T099	J35	1	625	16	739	evict	M01
1128	T091	J30	1	625	622	783	evict	M01
1128	T137	J52	9	6250	1590	70	start	M01
1128	T091	J30	1	625	622	783	start	M05
1128	T099	J35	1	625	16	739	start	M05
1128	T096	J32	0	1688	295	1070	start	M05
1128	T109	J42	0	1494	311	573	start	M05
1128	T123	J43	0	625	311	359	start	M08
1128	T124	J43	0	625	311	437	start	M08
1149	T136	J51	4	625	155	27	finish	M08
1158	T135	J50	0	6250	466	39	finish	M07
1162	T122	J43	0	625	311	334	finish	M08
1164	T115	J43	0	625	311	336	finish	M06
1165	T119	J43	0	625	311	337	finish	M08
1174	T111	J43	0	625	311	346	finish	M04

time	task	job	priority	CPU	memory	run time	event	machine
1174	T112	J43	0	625	311	346	finish	M05
1176	T116	J43	0	625	311	348	finish	M06
1176	T118	J43	0	625	311	348	finish	M08
1188	T110	J43	0	625	311	360	finish	M04
1196	T113	J43	0	625	311	368	finish	M05
1198	T137	J52	9	6250	1590	70	finish	M01

Cell window 26: Events

Time 544: machine M02 failed; task T059 stopped and went back to the queue.

Time 544: task T059 restarted on machine M06.

Time 639: task T090 (priority 9) evicted task T071 (priority 1) from machine M05.

Time 639: task T071 restarted on machine M04.

Time 1128: task T137 (priority 9) evicted task T091 (priority 1) from machine M01.

Time 1128: task T137 (priority 9) evicted task T096 (priority 0) from machine M01.

Time 1128: task T137 (priority 9) evicted task T099 (priority 1) from machine M01.

Time 1128: task T137 (priority 9) evicted task T109 (priority 0) from machine M01.

Time 1128: task T137 (priority 9) evicted task T123 (priority 0) from machine M01.

Time 1128: task T137 (priority 9) evicted task T124 (priority 0) from machine M01.

Time 1128: task T091 restarted on machine M05.

Time 1128: task T096 restarted on machine M05.

Time 1128: task T099 restarted on machine M05.

Time 1128: task T109 restarted on machine M05.

Time 1128: task T123 restarted on machine M08.

Time 1128: task T124 restarted on machine M08.

Longest waits (submission to first start, tasks started by the window end): task T092 waited 106 s (submitted at 649, first started at 755); task T096 waited 66 s (submitted at 658, first started at 724); task T085 waited 17 s (submitted at 565, first started at 582); task T086 waited 17 s (submitted at 565, first started at 582); task T087 waited 17 s (submitted at 565, first started at 582).

Cell window 26: Queue at window end

State after the events at time 1200.

Tasks running at time 1200 (remaining run time = seconds still to run after time 1200):

task	job	priority	CPU	memory	machine	started	remaining
T100	J36	1	4376	777	M03	780	214
T128	J46	0	1688	295	M03	908	773
T089	J28	1	1250	777	M04	609	548
T125	J43	0	625	311	M04	836	14
T126	J44	1	1250	777	M04	865	786
T127	J45	8	1250	2863	M04	894	2581
T130	J48	8	625	155	M04	1093	106
T091	J30	1	625	622	M05	1128	711
T096	J32	0	1688	295	M05	1128	998
T099	J35	1	625	16	M05	1128	667
T109	J42	0	1494	311	M05	1128	501
T092	J31	4	6873	1910	M06	755	2486

task	job	priority	CPU	memory	machine	started	remaining
T103	J39	1	3125	777	M06	825	201
T114	J43	0	625	311	M06	828	37
T117	J43	0	625	311	M06	828	201
T120	J43	0	625	311	M08	828	46
T121	J43	0	625	311	M08	828	3
T123	J43	0	625	311	M08	1128	287
T124	J43	0	625	311	M08	1128	365

Tasks waiting at time 1200, in queue order (each will run its full run time once started):

task	job	priority	CPU	memory	submitted	run time
T093	J31	4	6873	1910	649	2911
T094	J31	4	6873	1910	649	2994
T095	J31	4	6873	1910	649	2907
T105	J41	4	6873	1910	827	2936
T106	J41	4	6873	1910	827	2954
T107	J41	4	6873	1910	827	3065
T108	J41	4	6873	1910	827	2937
T131	J49	4	6873	1910	1108	3002
T132	J49	4	6873	1910	1108	2924
T133	J49	4	6873	1910	1108	2901
T134	J49	4	6873	1910	1108	2924

Machine M02 is down at time 1200; it returns at time 1207.